

QuantSight

An illustrated guide to the complete system

Features, data, models,
explainable AI and backtesting

Simple explanations beside real screenshots.
Includes the workflow from datasets to deployment.

Prepared 5 October 2026

Code baseline: 8bd515a | Screenshots: local running system

The screenshot displays the QuantSight Research Workspace interface. On the left is a sidebar with navigation links: Dashboard, Market, AI Compare, Leaderboard, Track Record, Backtesting, Portfolio, Game, and Settings. Below these is a 'Sign In' button and a section for 'Research & learning' with 'AI Decision Support'. The main content area is titled 'QUANTSIGHT RESEARCH WORKSPACE' and includes a 'Five-session prediction horizon' filter. A large heading reads 'Understand the signal. Check the evidence.' with a subtext: 'Explore model predictions, compare their historical performance, and follow outcomes as new forecasts mature.' Below this are two buttons: 'Explore backtesting' and 'Open market explorer'. A section titled 'Historical model snapshot' states 'Saved model outputs for research; these are not current trading signals.' and includes a date filter 'Signal dates: 2024-12-20'. This section contains four cards: 'Instruments covered' (44), 'Historical BUY votes' (4), 'Unanimous model votes' (10), and 'Mean model confidence' (51.6%). Below these are three more cards: 'Historical simulation' (Backtesting), 'Controlled research' (AI Compare), and 'Forward evaluation' (Track Record). At the bottom, there are two sections: 'SAVED CONSENSUS' with 'Historical BUY candidates' and 'LANGUAGE ANALYSIS' with 'Saved news sentiment'. A chat bubble icon is in the bottom right corner.

QuantSight

Dashboard

Market

AI Compare

Leaderboard

Track Record

Backtesting

Portfolio

Game

Settings

Sign In

Research & learning
AI Decision Support

QUANTSIGHT RESEARCH WORKSPACE

Five-session prediction horizon

Understand the signal. Check the evidence.

Explore model predictions, compare their historical performance, and follow outcomes as new forecasts mature.

Explore backtesting

Open market explorer

Historical model snapshot

Saved model outputs for research; these are not current trading signals.

Signal dates: 2024-12-20

Instruments covered 44 Stocks and ETFs in the saved snapshot	Historical BUY votes 4 17 SELL / 23 HOLD	Unanimous model votes 10 All available models agree on direction	Mean model confidence 51.6% Average model score; not ensemble accuracy
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Historical simulation Backtesting → Compare net returns, Sharpe, drawdown and trading costs against benchmarks.	Controlled research AI Compare → Inspect finance, VADER and FinBERT results alongside simple baselines.	Forward evaluation Track Record → Follow recorded forecasts and their outcomes after five trading sessions.
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SAVED CONSENSUS

Historical BUY candidates

LANGUAGE ANALYSIS

Saved news sentiment

44 instruments / 5-session direction target / 15 backtest portfolios

Find the feature, then follow the evidence

01 / Dashboard and market

[Pages 6-11](#)

02 / Stock analysis and explainable AI

[Pages 12-30](#)

03 / Model comparison and research

[Pages 31-41](#)

04 / Historical backtesting

[Pages 42-50](#)

05 / Live evidence and user tools

[Pages 51-63](#)

06 / Deployment, metrics and demo

[Pages 64-70](#)

Before the feature pages

- Page 3: dataset-to-deployment graphic tree.
- Page 4: datasets, cleaning and prediction target.
- Page 5: model families and training methods.

After the feature pages

- Deployment and service connections.
- Backtest execution and metric cheat sheet.
- Current limitations and a demo route.
- Evidence references and reproducibility.

Each feature page explains what it is, its data source, its calculation and how to interpret it.

Graphic tree: from datasets to deployment



Dashed branch: Supabase SQL migrations are applied separately; a git push does not run SQL.

What enters the system, and what changes

Saved finance panel	104,076 rows / 44 instruments Historical models, charts and backtests	2015-05-01 to 2024-12-20
GDELT daily text panel	2,127 daily rows Saved news display; no training overlap	2026-01-22 to 2026-04-23
WSB text and emotion	11,213 posts per scored source Sparse early text coverage; none in 2023-2024	Ends 2021-08-31
Research news archive	70,974 source rows; 26 tickers covered Separate controlled VADER / FinBERT study	Timestamped historical coverage
Live Yahoo daily panel	Required trained universe + sector data Single XGBoost finance inference	Latest eligible completed NYSE session

Preparation in plain words

Check dates, duplicates, gaps and valid prices. Calculate RSI, MACD, bands, averages, ranges, volume ratios and relative-market inputs using trailing data. Keep raw prices for display; trained price-level inputs use relative values.

The label to predict

UP: Close five sessions later is higher than the signal-session Close. DOWN: equal or lower. Unknown future tails stay missing in new labels. Keep text availability flags before filling missing numeric values; old text joins still have timing limitations.

What the prediction models do

XGBoost

Boosted decision trees for daily tabular inputs. The finance bundle is also used for live inference.

Random Forest

Many decision trees vote together; useful as a different tree-based comparison.

Logistic Regression

A simple linear probability model; a baseline against more complex methods.

LSTM with attention

A recurrent sequence model using ten sessions. Attention weights its input days.

GRU

Another recurrent sequence model with a simpler gated state than LSTM.

Transformer Encoder

A separate self-attention sequence model. It is not the legacy LSTM+Transformer label.

A model outputs a five-session direction score. It does not predict a guaranteed price, execute orders or retrain itself when a page opens.

[Code / evidence: backend/train_xgboost.py](#)

Two input variants

Each family has finance-only and finance + sentiment/emotion versions: 12 saved variants. The ensemble averages the four core XGBoost/LSTM probabilities.

Time-ordered training

Legacy training uses chronological train/test dates and purged walk-forward checks. Scalers use training data only. XGBoost uses training out-of-fold outputs for isotonic or Platt calibration.

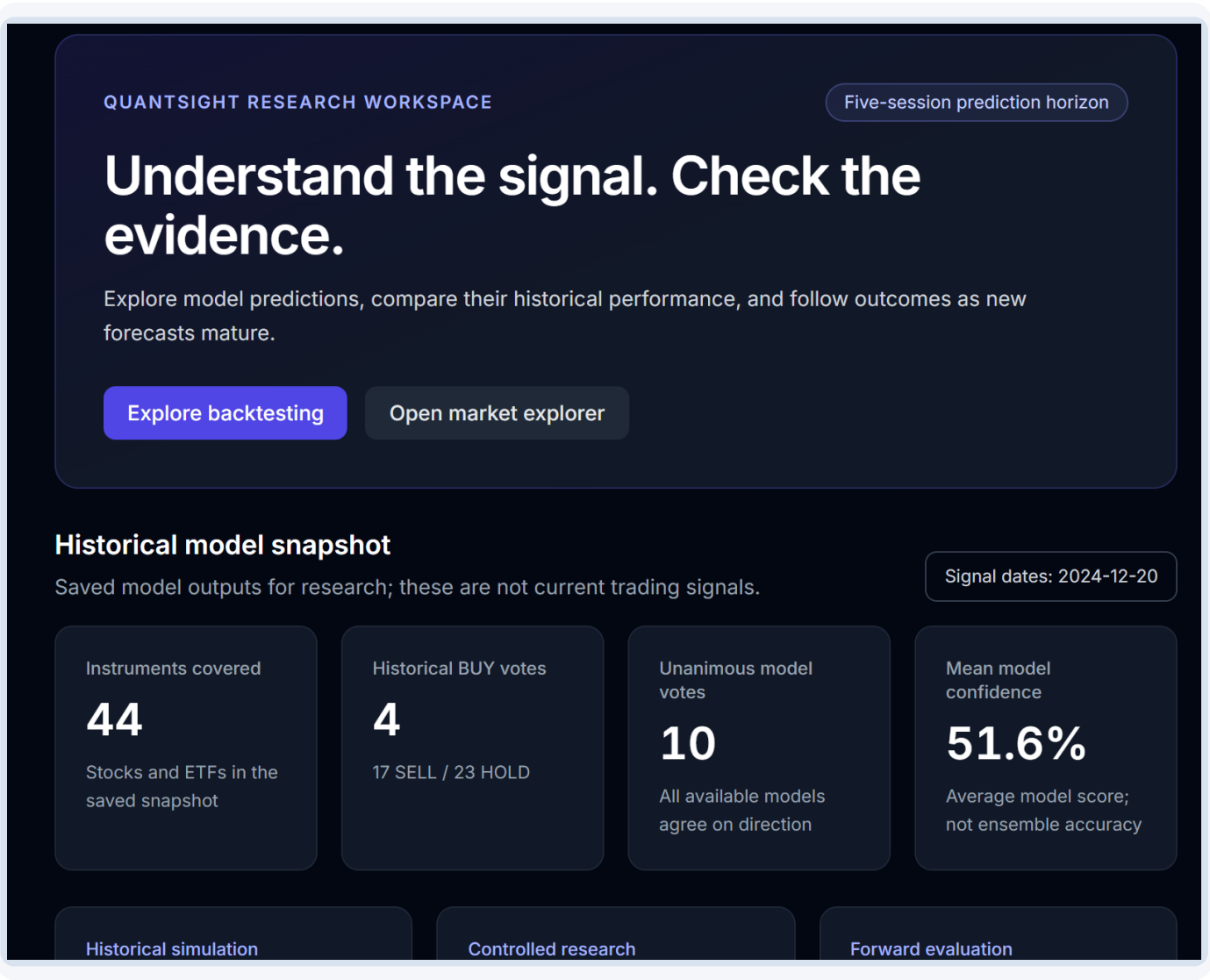
Saved output contract

Historical confidence is calibrated P(UP) in percent. Legacy direction uses the raw score. Live direction confidence converts P(UP) to the selected direction.

Research stays separate

The newer controlled finance study fits three fixed models on 13 ratio features. Its results do not replace the legacy serving bundle. The candidate 2025 test has not been scored here.

Dashboard: the saved model snapshot



WHAT IT IS

The home page gives a quick view of the research system and links to its evidence.

WHERE DATA COMES FROM

The API reads the latest saved prediction row for each of 44 stocks and ETFs. The displayed signal date is 20 December 2024.

HOW THE RESULT IS MADE

The four core models vote on direction. More BUY votes give BUY; more SELL votes give SELL; a tie gives HOLD. The cards count these outcomes.

HOW TO READ IT

Mean confidence is an average model score. It is not the measured accuracy of the ensemble or a current market signal.

Dashboard: historical BUY candidates

SAVED CONSENSUS

Historical BUY candidates

Ranked by model agreement, then confidence.
Open an instrument to inspect its saved analysis.

1

DIA

SPDR Dow Jones Industrial Average ...

Strong agreement

BUY

53.1%

2

TLT

iShares 20+ Year Treasury Bond ETF

Strong agreement

BUY

52.9%

3

AAPL

Apple Inc.

Moderate agreement

BUY

53.1%

4

MRNA

Moderna Inc.

Moderate agreement

BUY

53.1%

Confidence and agreement are model outputs, not a
measured probability of investment success.

WHAT IT IS

This list helps the user find instruments to inspect more closely.

WHERE DATA COMES FROM

The same dated, four-model prediction snapshot supplies the candidates and their individual scores.

HOW THE RESULT IS MADE

The page keeps overall BUY results, then ranks them by model agreement and confidence. Selecting a ticker opens Stock Detail.

HOW TO READ IT

Agreement means the models chose the same direction. Models can agree and still be wrong. These are saved research outputs.

Dashboard: saved news sentiment

LANGUAGE ANALYSIS

Saved news sentiment

VADER scores headline language. These scores are separate from the historical prediction snapshot.

Archive window: 2026-04-09 to 2026-04-23

Latest saved article: 164 days old. This is not a live news feed.

Positive	Neutral	Negative
35.5%	49.9%	14.6%

1,032 unique headlines in a 14-day archive window.

► Daily sentiment values

WHAT IT IS

Positive, neutral and negative shares describe the language in the saved headlines.

WHERE DATA COMES FROM

The GDELT archive is scored using VADER. This view covers 1,032 unique headlines in its recent saved 14-day window, ending 23 April 2026.

HOW THE RESULT IS MADE

The server deduplicates archived headlines, groups their polarity labels and calculates each group as a share of the total. Daily values are expandable.

HOW TO READ IT

This is an archive, not a live feed. Its 2026 news is separate from the models' 2024 signal date. Language sentiment is not return accuracy.

Dashboard: archived headline links

Saved news headlines

Latest entries in the GDELT archive, with their publication dates.

PG / 2026-04-23 / baijiahao.baidu.com

How Microsoft spooked the global carbon removal market

MSFT / 2026-04-22 / moneycontrol.com

Berkshire Hathaway (TSE : BRK) Reaches New 52 - Week Low – Here What Happened

BRK-B / 2026-04-22 / dailypolitical.com

Separate from model signal dates

Danaher Announces Pricing of Euro - Denominated Senior Notes Offering

DHR / 2026-04-23 / prnewswire.com

Should You Buy Alphabet Stock Before April 29 ?

GOOGL / 2026-04-22 / finance.yahoo.com

These Analysts Increase Their Forecasts On UnitedHealth After Upbeat Earnings - UnitedHealth Group (NYSE :

UNH / 2026-04-22 / benzinga.com

Educational decision support. Historical simulations and new live outcomes are evaluated separately.

WHAT IT IS

The news list lets the user inspect the underlying headline, ticker, date and publisher.

WHERE DATA COMES FROM

These are stored GDELT entries returned by the market-news API. The page links to the original publishers.

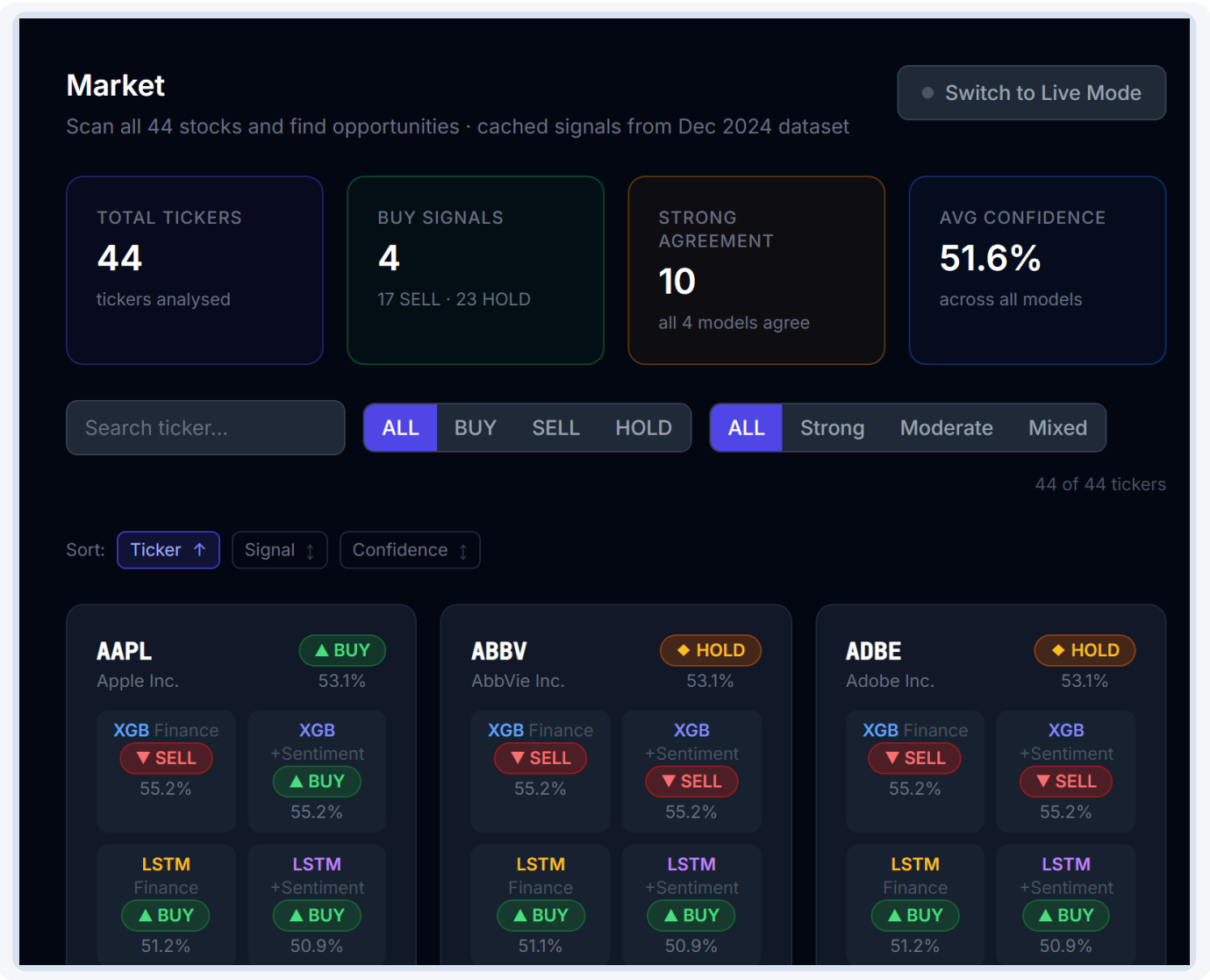
HOW THE RESULT IS MADE

The API sorts saved entries by date and returns a bounded list. Clicking a headline opens its publisher in a new tab.

HOW TO READ IT

A ticker match can be noisy, and publishers may change their pages. Saved headlines do not become inputs to a past prediction simply by appearing here.

Market: scan and filter instruments



WHAT IT IS

The market table compares the four core model signals for each instrument.

WHERE DATA COMES FROM

Historical mode reads the saved prediction CSVs and the explanation API. The default rows are from the December 2024 snapshot.

HOW THE RESULT IS MADE

Search by ticker; filter by BUY, SELL or HOLD and agreement; sort the results. Overall direction comes from model votes. Risk uses indicator rules.

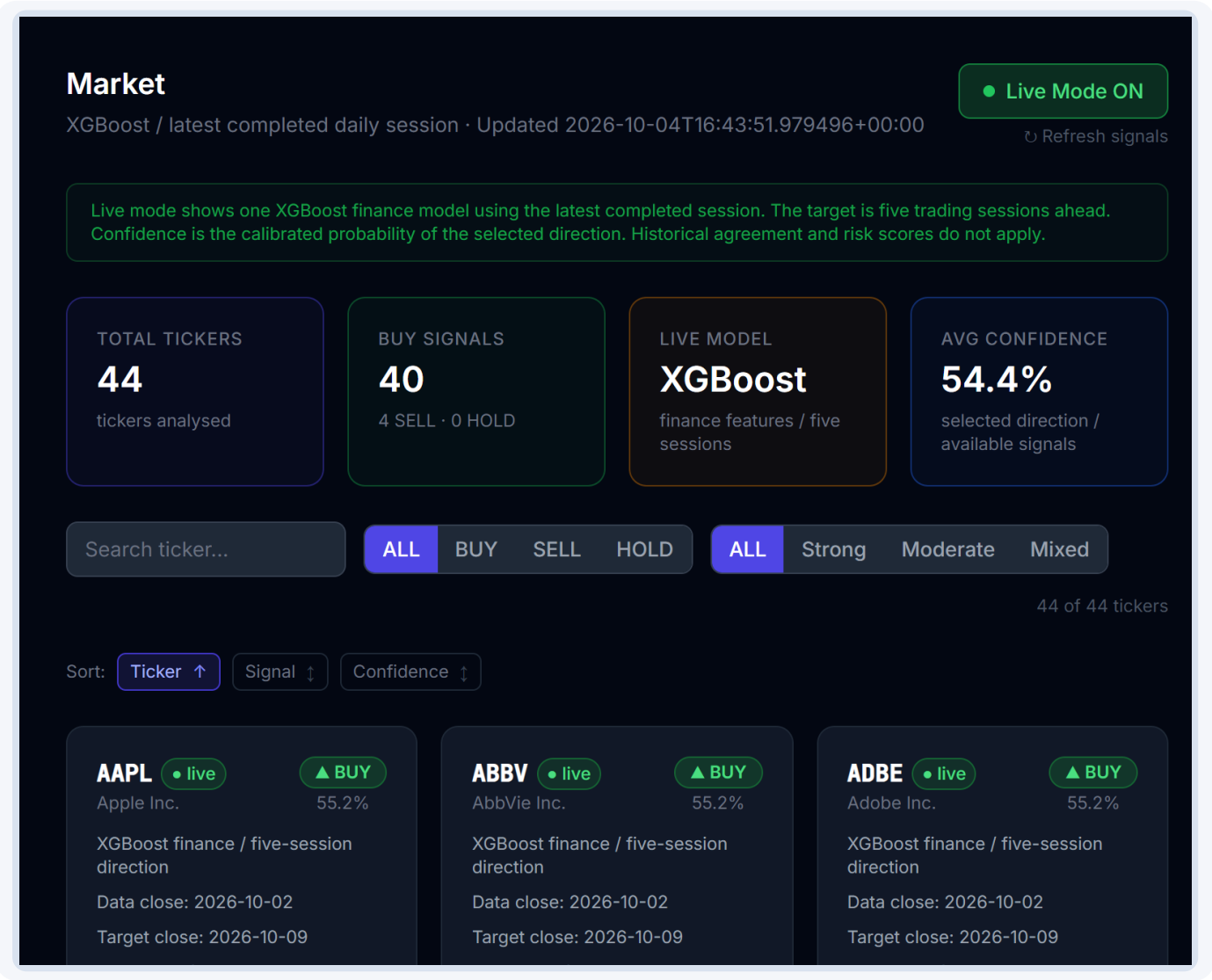
HOW TO READ IT

Historical confidence columns contain calibrated P(UP), including on SELL rows. The displayed risk categories are rules, not estimated loss probabilities.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: frontend/src/pages/Overview.jsx

Market: current daily inference



WHAT IT IS

Live mode requests a new finance-only XGBoost prediction using a completed daily market session.

WHERE DATA COMES FROM

Yahoo Finance supplies adjusted daily prices for the required instrument and sector universe. The backend applies the saved model bundle.

HOW THE RESULT IS MADE

It checks the NYSE calendar, excludes unfinished bars, recreates the 30 finance inputs, scales them and applies the saved calibrator. The target is five sessions later.

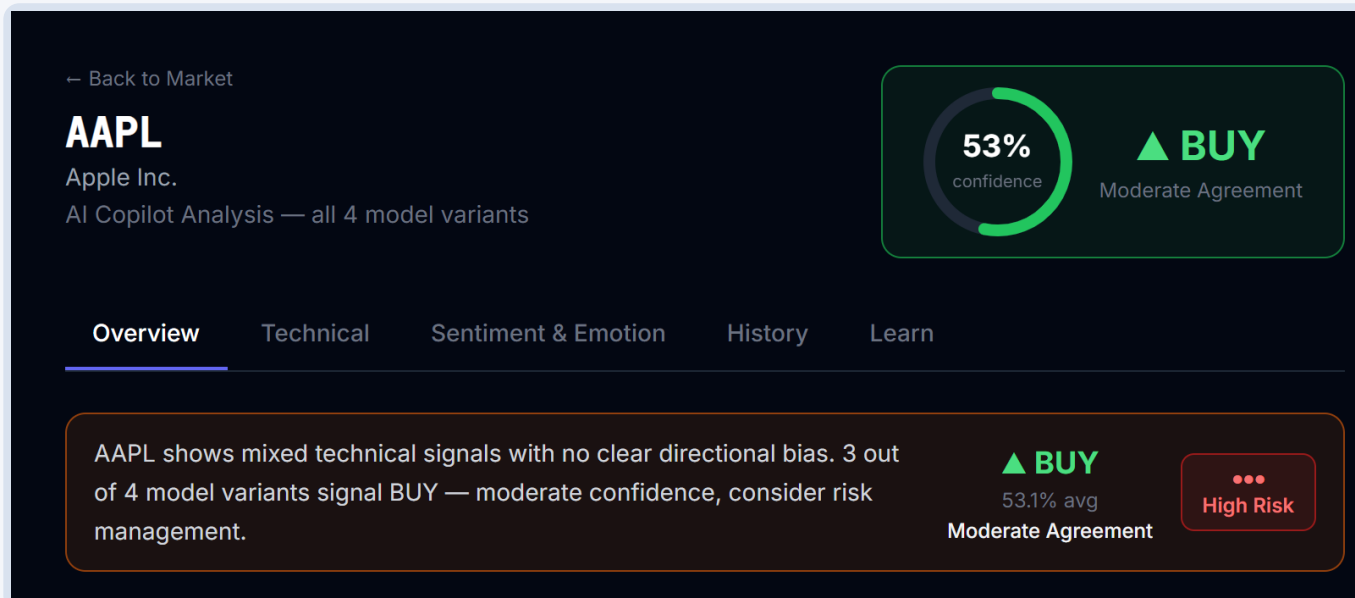
HOW TO READ IT

This is daily inference, not an intraday forecast. Missing or stale required inputs produce an unavailable result. Historical four-model votes are not used in live mode.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/live_signals.py](#)

Stock Detail: signal and navigation



WHAT IT IS

Stock Detail brings together the signal, charts, explanations, news, history and learning tools for one ticker.

WHERE DATA COMES FROM

The stock and explanation APIs read the saved price features and four core model outputs for AAPL in this example.

HOW THE RESULT IS MADE

The overview combines model votes and their average P(UP). Tabs switch between Overview, Technical, Sentiment & Emotion, History and Learn.

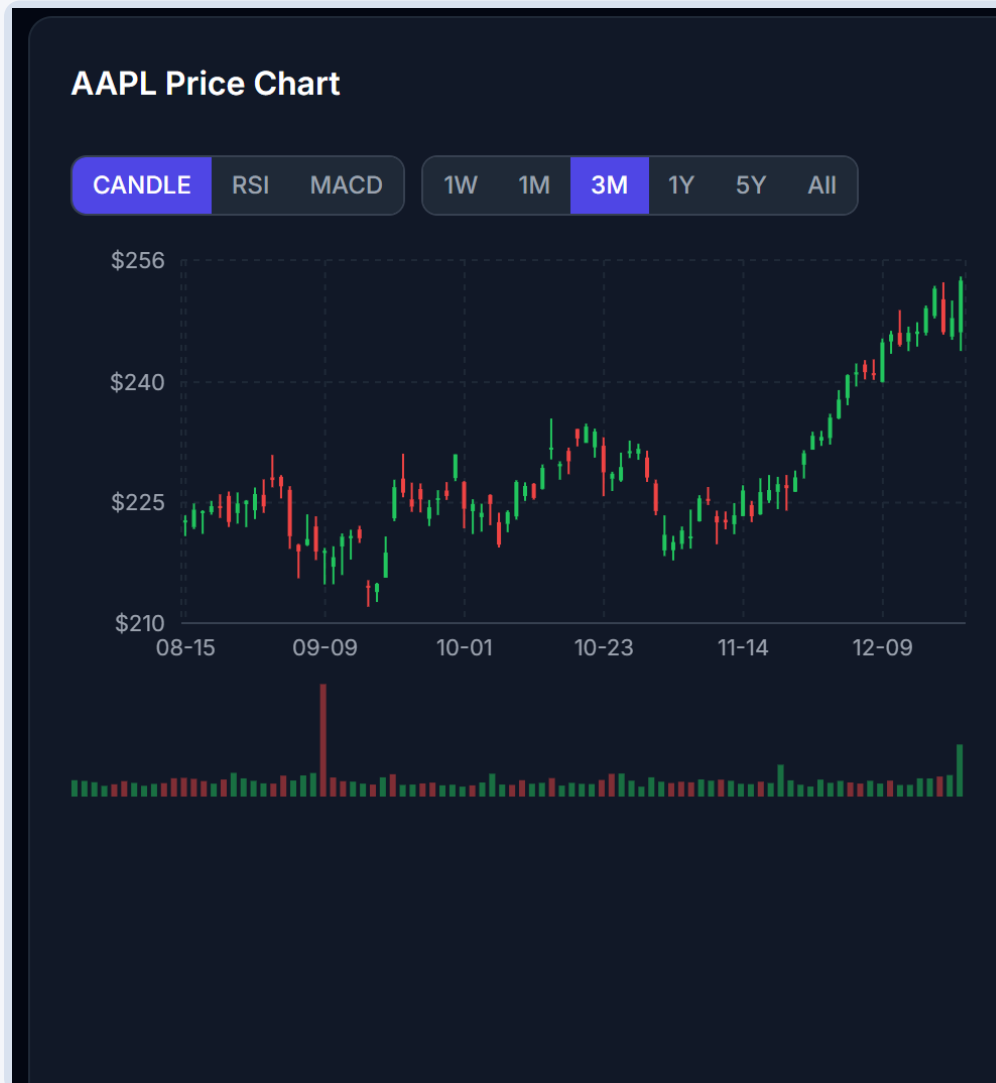
HOW TO READ IT

The overall vote can differ from one individual model. The headline confidence is an average score, not the probability that buying the stock will make money.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Detail.jsx](#)

Price chart: candles and volume



WHAT IT IS

Each candle shows one session's open, high, low and close. The bars below show volume.

WHERE DATA COMES FROM

The stock API reads the saved OHLCV price panel. Period controls request different lengths of this daily history.

HOW THE RESULT IS MADE

The candle body runs from open to close; the wick shows the high and low. Colour shows whether that day closed above or below its open.

HOW TO READ IT

Green candles are observed price moves, not BUY predictions. This historical chart ends in December 2024 and does not show intraday candles.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/components/CandlestickChart.jsx](#)

RSI chart and saved signal markers



WHAT IT IS

The RSI view compares recent upward and downward price moves on a scale from 0 to 100.

WHERE DATA COMES FROM

RSI values come from the saved feature panel. The chosen model's saved BUY/SELL dates supply the optional markers.

HOW THE RESULT IS MADE

RSI uses a 14-session lookback. Guide lines at 30 and 70 help identify low and high momentum readings; controls choose the period and model overlay.

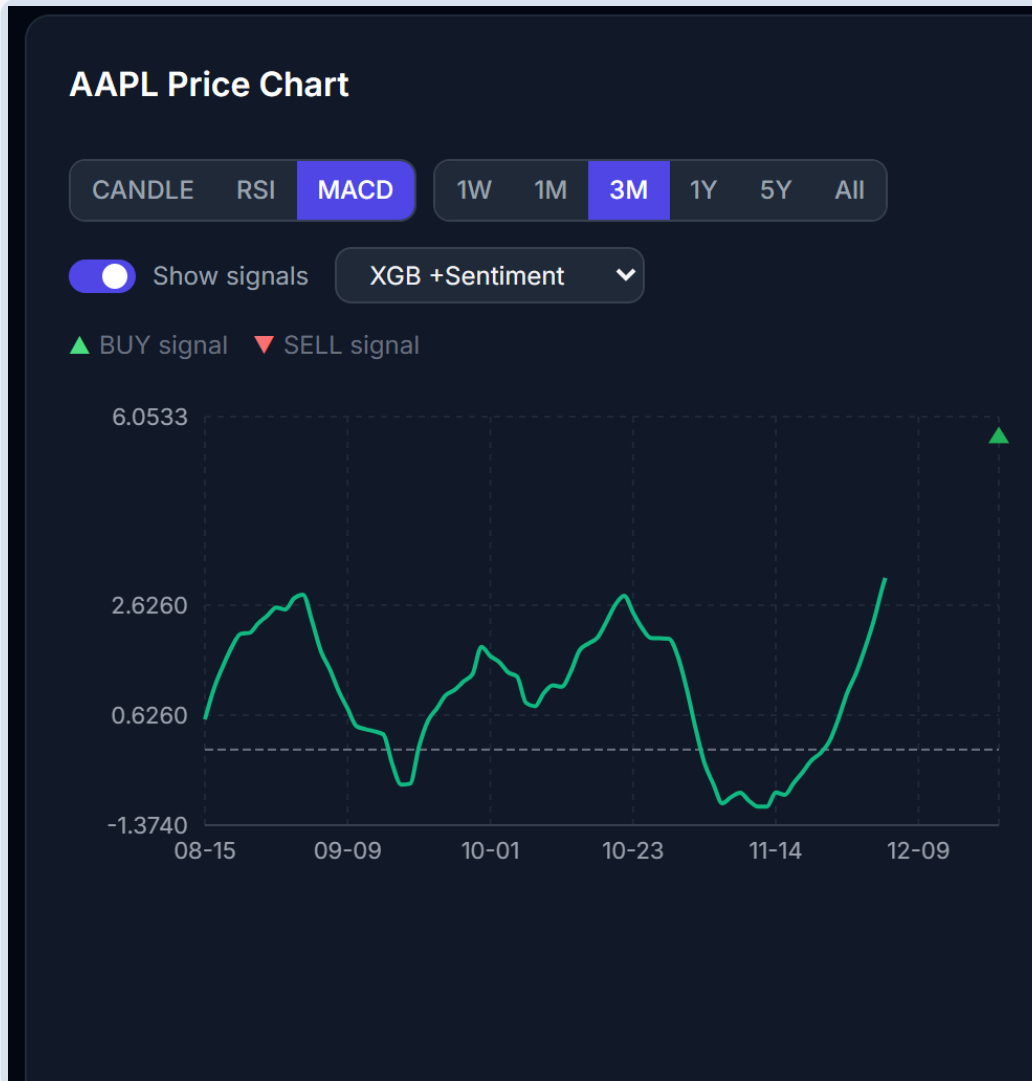
HOW TO READ IT

An RSI threshold is a rule of thumb. A high or low value does not guarantee a reversal, and markers are model decisions rather than executed trades.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Detail.jsx](#)

MACD chart and model overlay



WHAT IT IS

MACD shows the difference between shorter and longer exponential moving averages.

WHERE DATA COMES FROM

The saved price features provide MACD. Signal markers are read from the selected core model's historical predictions.

HOW THE RESULT IS MADE

MACD compares the 12- and 26-session averages. Its 9-session signal line is used by the indicator rules; the chart displays MACD over the selected period.

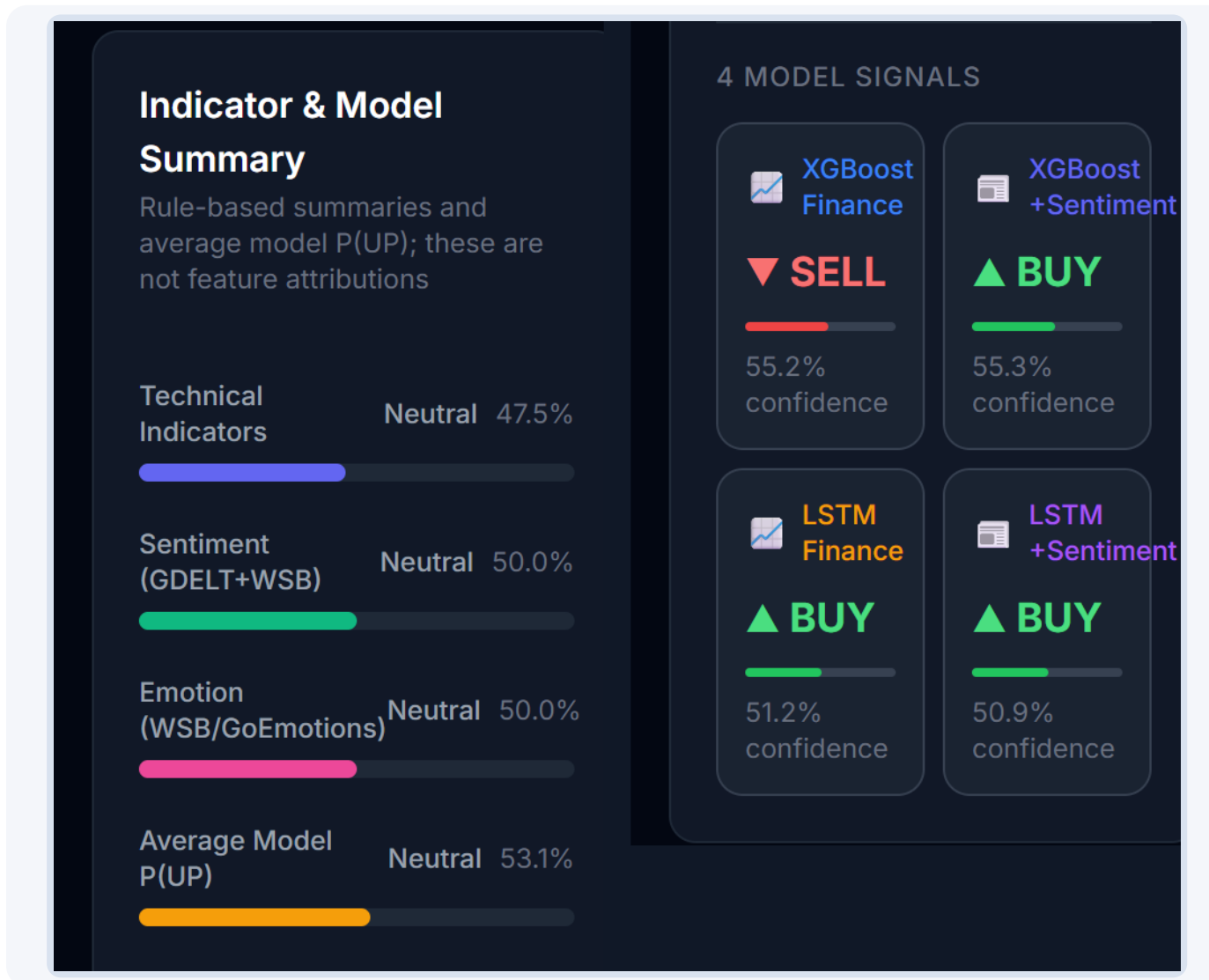
HOW TO READ IT

A bullish indicator reading does not force a model to predict BUY. The model considers several inputs, while the chart shows one indicator.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/technical_indicators.py](#)

Indicator scores and model votes



WHAT IT IS

The summary separates technical, sentiment and emotion assessments from the four saved model signals.

WHERE DATA COMES FROM

The explanation API reads the latest historical feature row and each core model's output.

HOW THE RESULT IS MADE

Indicator rules produce the summary scores. The model section shows XGBoost Finance, XGBoost +Sentiment, LSTM Finance and LSTM +Sentiment.

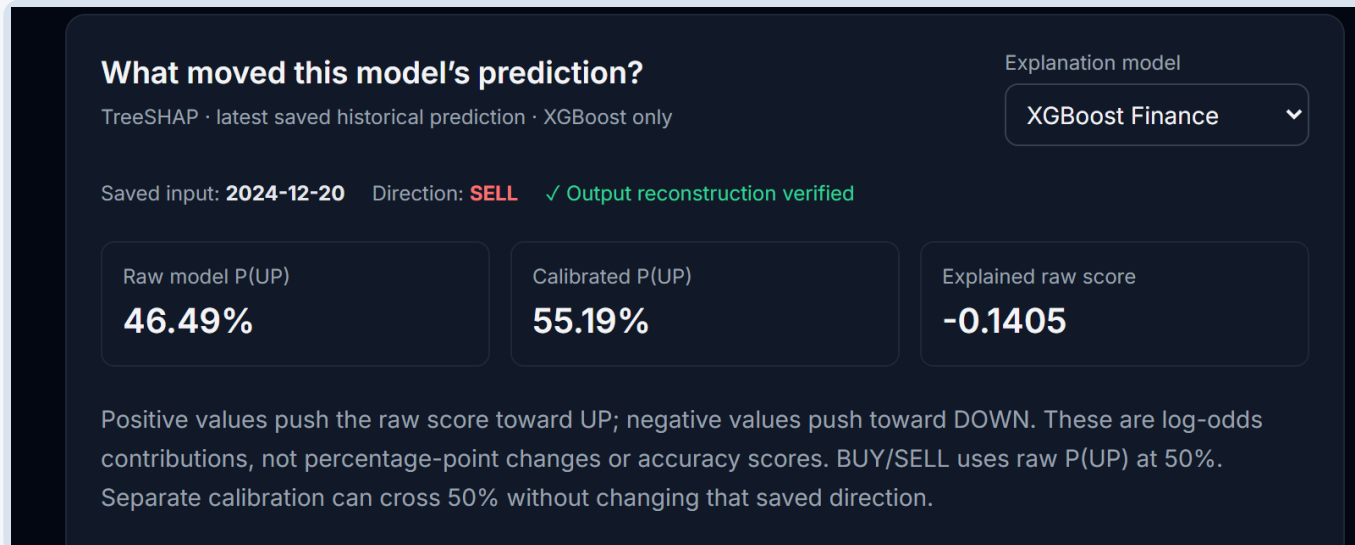
HOW TO READ IT

These indicator scores are not SHAP contributions. Historical text values are zero in the current test period. Differences between variants do not prove a news benefit.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

Explainable AI: the exact saved prediction



WHAT IT IS

The attribution panel explains one saved XGBoost output. Finance and Finance + Sentiment can be selected.

WHERE DATA COMES FROM

The backend loads the exact saved model bundle and ticker/date input row, then verifies the prediction against the saved CSV.

HOW THE RESULT IS MADE

The panel separates raw P(UP), calibrated P(UP) and the raw score. For AAPL Finance, raw P(UP) is 46.49%, calibrated P(UP) is 55.19%, and direction remains SELL.

HOW TO READ IT

Direction uses the raw 50% threshold. Calibration can cross 50% without changing that saved label. This panel explains a historical XGBoost output, not the live model or consensus.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/feature_attribution.py](#)

Explainable AI: feature contribution bars



WHAT IT IS

The bars show which inputs pushed the tree model's raw score toward UP or DOWN.

WHERE DATA COMES FROM

Native XGBoost TreeSHAP calculates contributions from the saved trees after applying the bundle's feature order and scaler.

HOW THE RESULT IS MADE

The chart shows the eight largest absolute contributions and combines the remaining inputs. Baseline plus all contributions reconstructs the raw log-odds score.

HOW TO READ IT

Contributions are log-odds, not percentage points or causal effects on stock prices. A zero-valued input can still have a contribution in a tree model.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/feature_attribution.py](#)

Explainable AI: inputs and verification

▼ All 30 inputs and verification details

Feature	Saved value	Scaled input	Contribution
market_volatility	0.0273523	-0.650422	-0.2059
cmf_20	0.390019	2.43621	+0.1517
rsi_rel_market	27.4053	2.62009	+0.0917
clv	0.890442	1.41109	-0.0887
volume_sma_10	5.60906e+7	0.211449	-0.0733
macd_signal_norm	0.0200021	0.714488	-0.0622
atr	4.25759	0.662151	-0.0462
mfi_14	74.0209	1.36146	-0.0461
return5_rel_spy	0.0439539	1.02544	+0.0411
svi_10	0.636374	1.82169	+0.0344
volchg_rel_market	0.571079	0.000897543	-0.0335
macd_norm	0.0220067	0.747987	-0.0327
return_5	0.0256318	0.447601	+0.0310
sma_50_norm	-0.0776828	-0.748678	-0.0307
volatility_20	6.89684	0.985752	-0.0268
bb_lower_norm	-0.0966720	-0.468250	-0.0224
sma_10_norm	-0.0200205	-0.493244	+0.0206

WHAT IT IS

The expanded table lets the user inspect the actual input values, scaled values and contributions.

WHERE DATA COMES FROM

The table is built from the exact row used for this saved prediction. It contains all 30 finance inputs; the screenshot shows its first section.

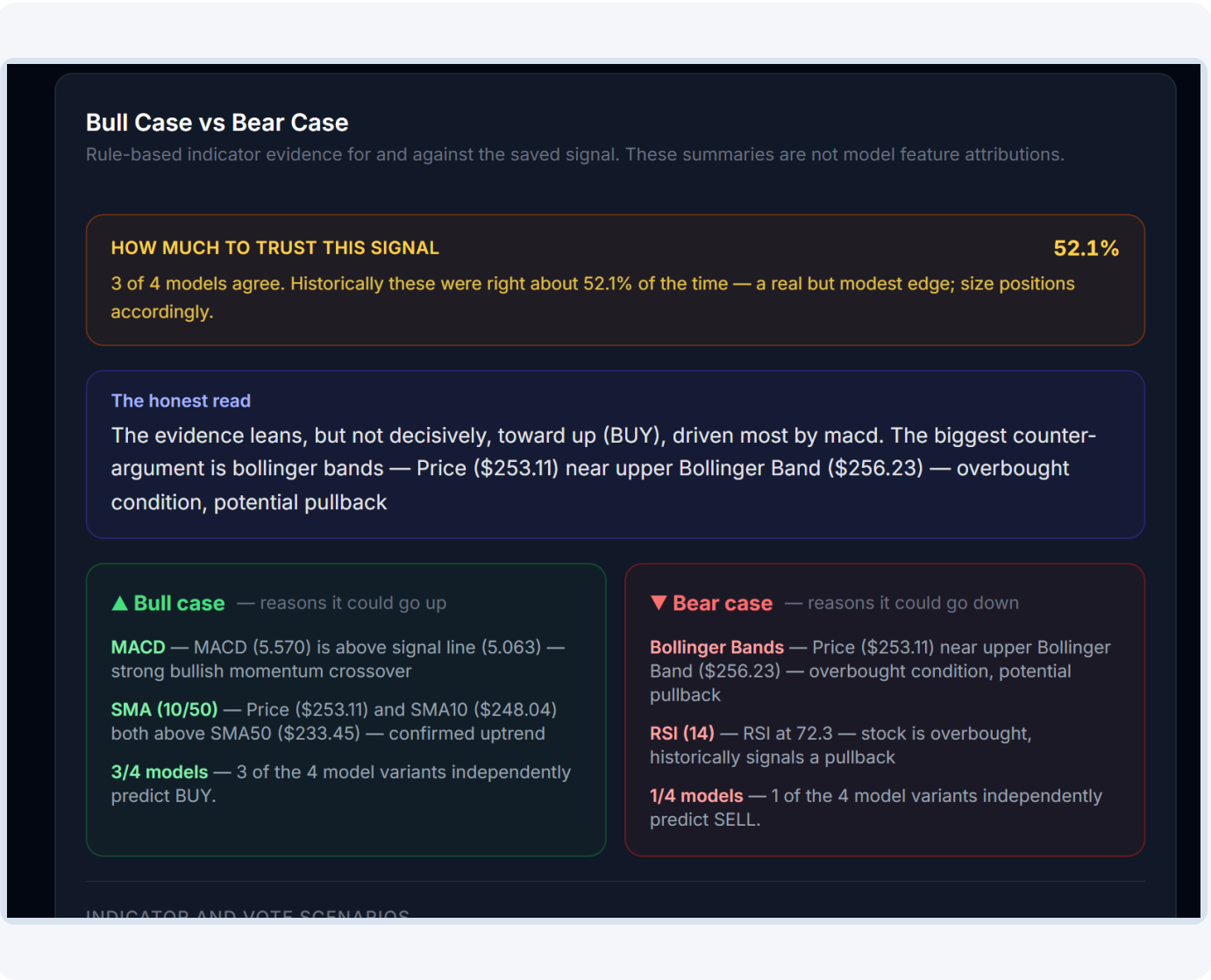
HOW THE RESULT IS MADE

The backend checks raw-score reconstruction, raw probability, calibrated probability and saved direction. Model and input hashes identify the evidence version.

HOW TO READ IT

A verification failure produces an unavailable explanation. Matching the saved prediction proves consistency, not that the prediction is correct or that the model has a fresh trading edge.

Bull case and bear case



WHAT IT IS

This panel presents evidence for both directions and scenarios that could change an indicator assessment or model vote count.

WHERE DATA COMES FROM

The explanation engine combines saved RSI, MACD, price trend, historical accuracy and the core models' votes.

HOW THE RESULT IS MADE

Rules describe supportive and opposing indicators. Reliability uses historical performance. The scenario section shows relevant indicator thresholds or vote changes.

HOW TO READ IT

These are rule-based summaries. Scenarios do not rerun the trained predictor or guarantee that its forecast would change. Historical reliability may not carry into a new period.

Indicator and vote scenarios

INDICATOR AND VOTE SCENARIOS

These threshold scenarios do not rerun the model or guarantee a prediction change.

RSI RSI is 72 — above 70, the classic overbought line. If it slipped back under 70, the pullback warning would clear.

MACD MACD (5.57) is above its signal line (5.06) — a bullish crossover. A move of 0.51 the other way would flip momentum bearish.

Model votes 3 of 4 models say BUY. If 1 flipped to SELL, the consensus would drop to a HOLD.

WHAT IT IS

Scenarios describe a threshold or vote change that would alter an indicator assessment or the overall model consensus.

WHERE DATA COMES FROM

The explanation engine uses the saved indicator values and the four models' direction votes.

HOW THE RESULT IS MADE

The displayed conditions identify a change to RSI, momentum, trend or the voting balance. They give the user a simple way to inspect sensitivity.

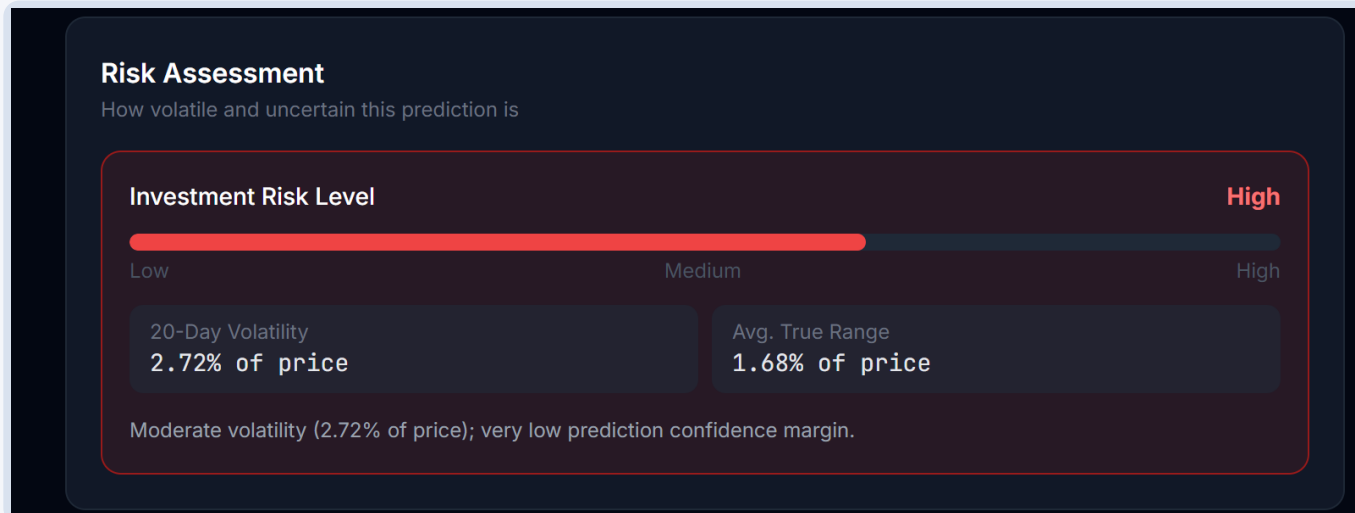
HOW TO READ IT

These conditions do not rerun the trained predictor. They are not verified counterfactual predictions and do not guarantee a change in its raw or calibrated score.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

Risk assessment



WHAT IT IS

The risk card gives a simple warning level for the current saved stock analysis.

WHERE DATA COMES FROM

It uses saved price indicators, model disagreement and confidence from the explanation engine.

HOW THE RESULT IS MADE

A fixed rule-based score maps the evidence to Low, Medium or High and lists contributing factors. It helps the user inspect uncertainty.

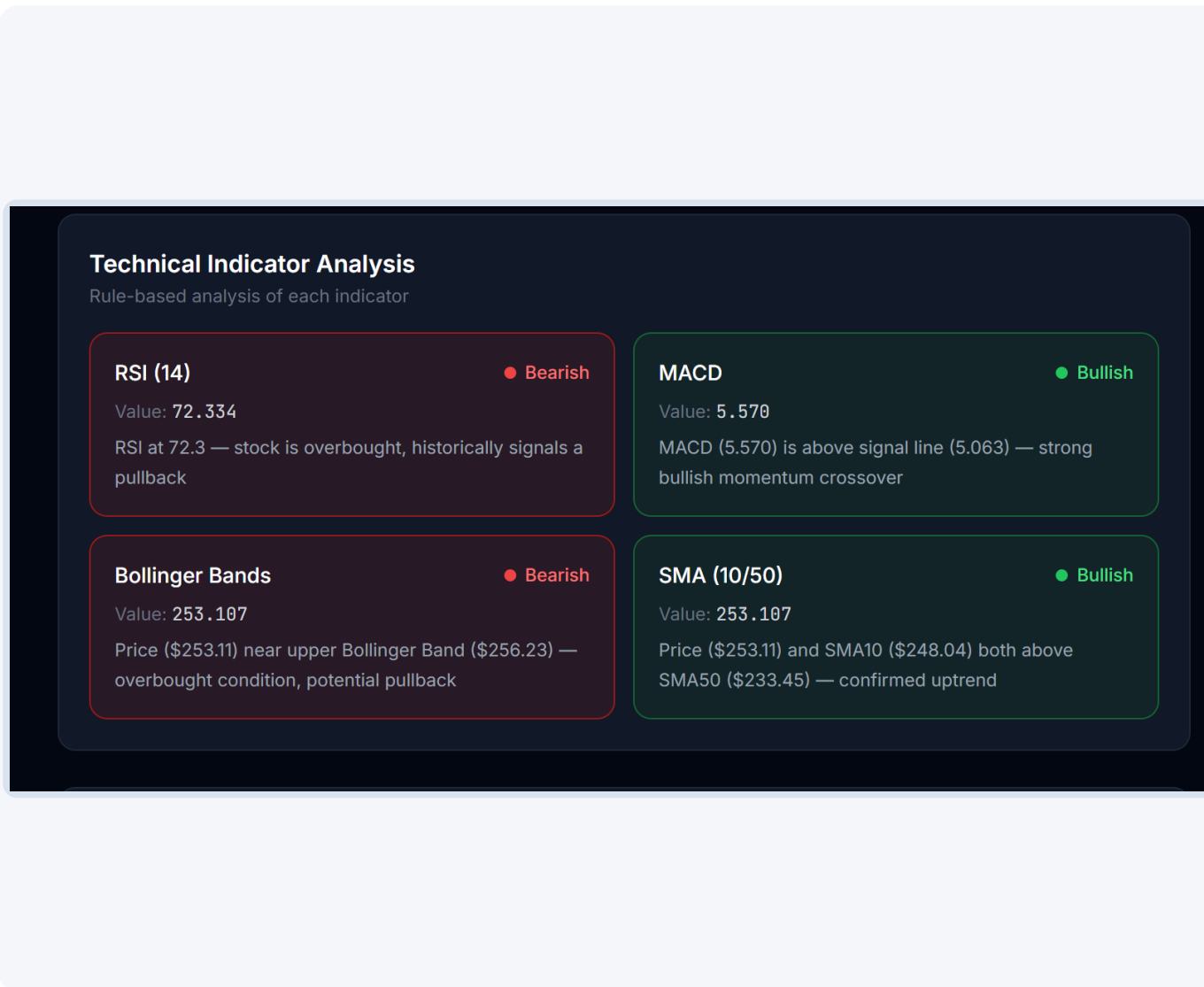
HOW TO READ IT

This is not portfolio VaR, a probability of losing money or a personalised suitability check. Backtesting has separate measured volatility and drawdown metrics.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

Technical tab: indicator explanations



WHAT IT IS

The technical cards describe RSI, MACD, Bollinger Bands and the moving-average trend in simple terms.

WHERE DATA COMES FROM

The saved finance feature row supplies each numeric value. The explanation engine supplies the accompanying text.

HOW THE RESULT IS MADE

Rules compare RSI with thresholds, MACD with its signal line, price with its bands, and price/SMA10 with SMA50.

HOW TO READ IT

The words Bullish and Bearish describe indicator rules. They are not independent trained models and do not guarantee the stock's next move.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

Technical tab: LSTM attention



WHAT IT IS

Attention weights show how the LSTM layer distributes weight across its ten input sessions.

WHERE DATA COMES FROM

These weights are saved with the historical LSTM prediction and returned by the stock-analysis data.

HOW THE RESULT IS MADE

The ten weights add to approximately 100%. The chart scales bars relative to one another so small differences are visible.

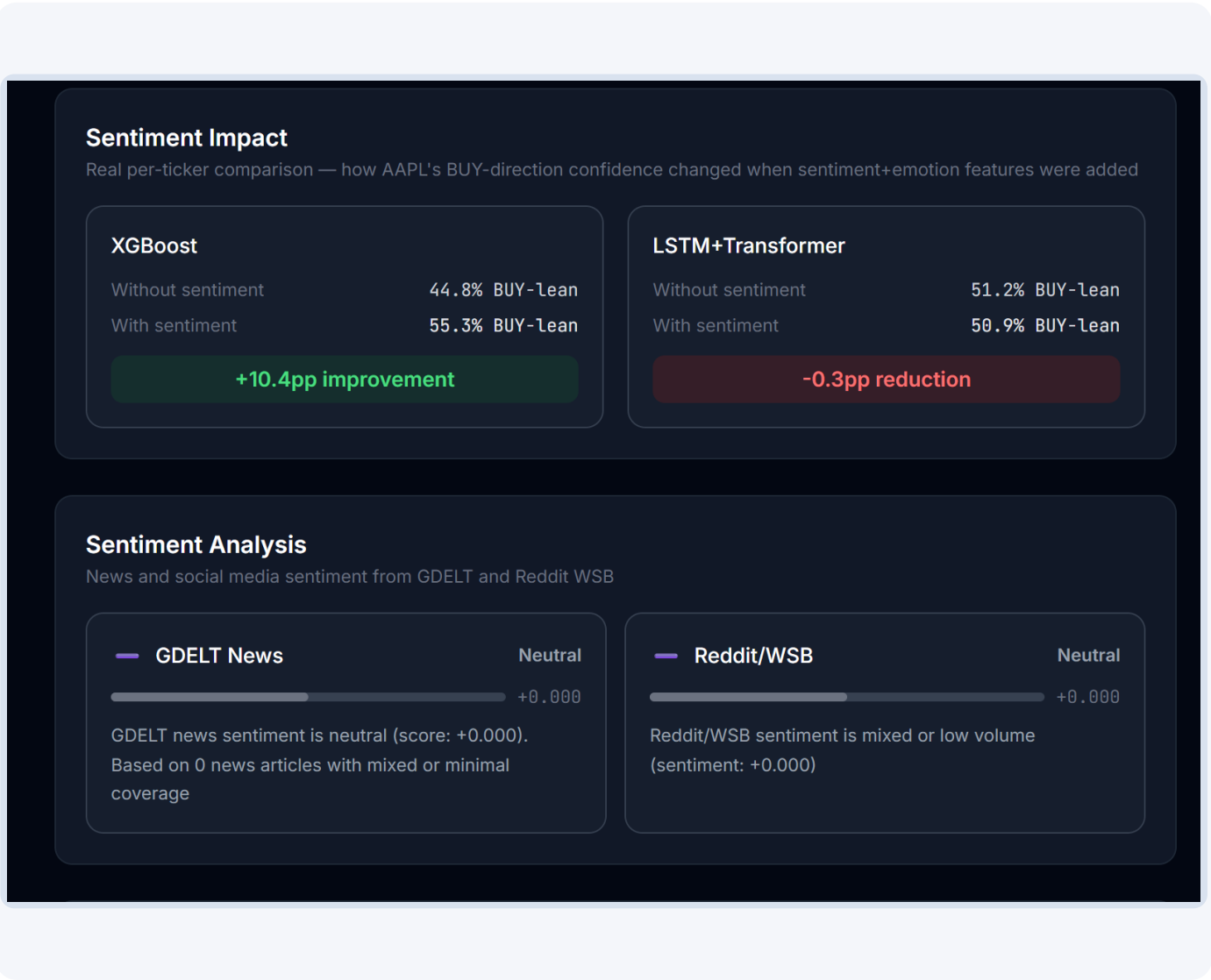
HOW TO READ IT

For AAPL, all weights are near the uniform 10% level. Attention is internal weighting, not exact reasoning, causal importance or TreeSHAP.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Detail.jsx](#)

Sentiment tab: language scores and variants



WHAT IT IS

This view separates saved news and social sentiment from the differences between finance-only and text-feature models.

WHERE DATA COMES FROM

GDELT and Reddit/WSB supply the original text sources. VADER scores language polarity, and daily aggregates are joined to the feature panel.

HOW THE RESULT IS MADE

The page displays available polarity scores and compares the saved model variants' P(UP) values. A difference is a descriptive model-output difference.

HOW TO READ IT

Current historical text inputs are zero on the latest prediction date. Labels such as improvement or impact in this older view should not be interpreted as proven sentiment benefit.

Sentiment tab: emotion features

Emotion Impact

Multi-label emotion detected in Reddit/WSB posts (GoEmotions) — distinct from sentiment polarity above

No WSB posts mentioning AAPL on the latest prediction date — emotion is only available on days with Reddit coverage.

WHAT IT IS

Emotion features describe feelings expressed in social posts, beyond positive or negative sentiment.

WHERE DATA COMES FROM

A pretrained GoEmotions-based RoBERTa model scores WSB posts. The project aggregates six groups: fear, optimism, anger, excitement, confusion and disappointment.

HOW THE RESULT IS MADE

Post scores are aggregated by ticker and day for the text-feature variants. The interface shows no emotion chart when the latest date has no WSB coverage.

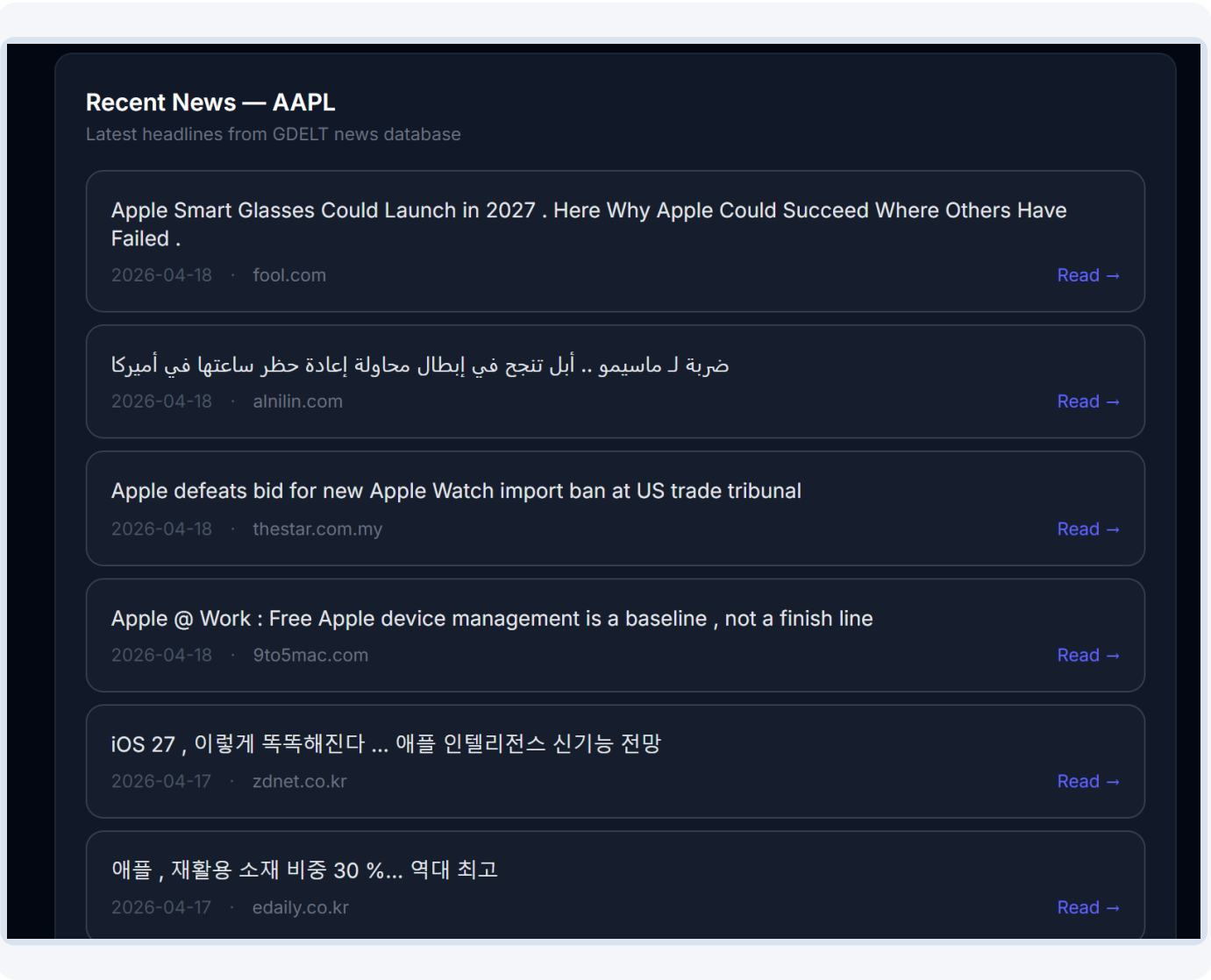
HOW TO READ IT

AAPL has no matching WSB posts on its latest saved prediction date. Missing coverage is not evidence that traders felt neutral, and emotion-model F1 is not stock prediction accuracy.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/build_features.py](#)

Sentiment tab: ticker-specific news



REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/main.py](#)

WHAT IT IS

Stock Detail also lists saved headlines associated with the selected ticker.

WHERE DATA COMES FROM

The news API reads the GDELT archive and returns a limited set of ticker matches, dates and publisher links.

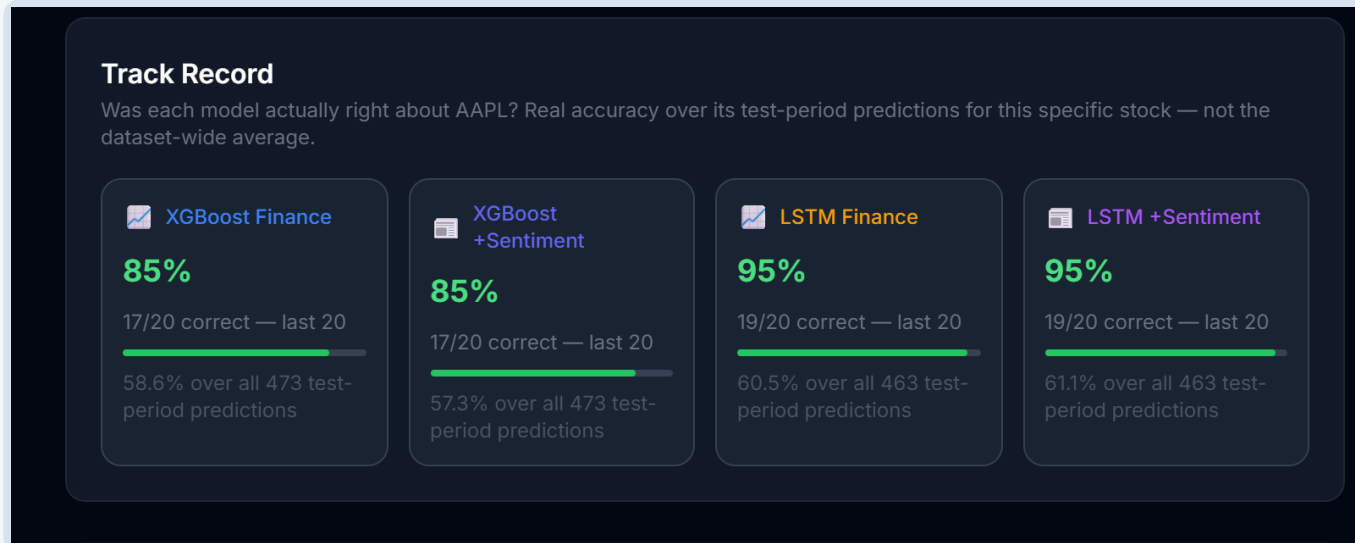
HOW THE RESULT IS MADE

Entries are filtered by ticker and ordered by saved date. The user can open a publisher to check the full context.

HOW TO READ IT

The AAPL headlines shown here are from April 2026. They cannot explain a December 2024 forecast, and multilingual or ambiguous matches require care.

History tab: per-stock accuracy



WHAT IT IS

This shows how often each core model matched the historical five-session outcome for this particular stock.

WHERE DATA COMES FROM

The accuracy-history API compares the saved prediction labels with the stored historical target labels for that ticker.

HOW THE RESULT IS MADE

Correct divided by evaluated predictions gives accuracy. The page shows a recent 20-prediction sample as well as each model's longer saved history.

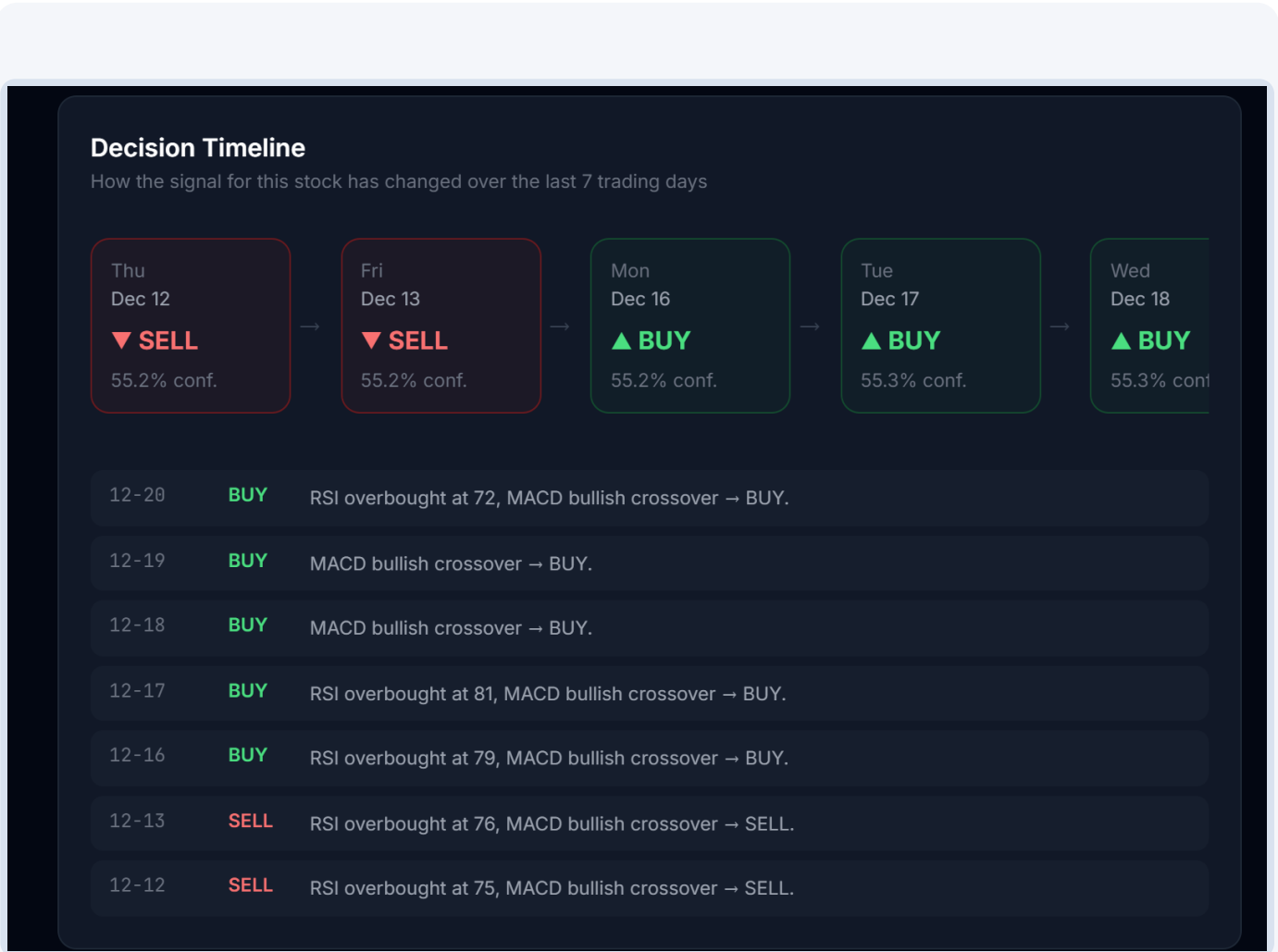
HOW TO READ IT

A small recent sample can look unusually strong. This reused historical window is separate from the forward Live Track Record, and overlapping targets are dependent.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

History tab: decision timeline



WHAT IT IS

The timeline shows how the saved signal changed over the last seven available sessions.

WHERE DATA COMES FROM

The history API reads dated core-model outputs and the corresponding historical indicator rows.

HOW THE RESULT IS MADE

It displays one dated signal card per session and a short indicator summary. The user can compare changes in direction and scores.

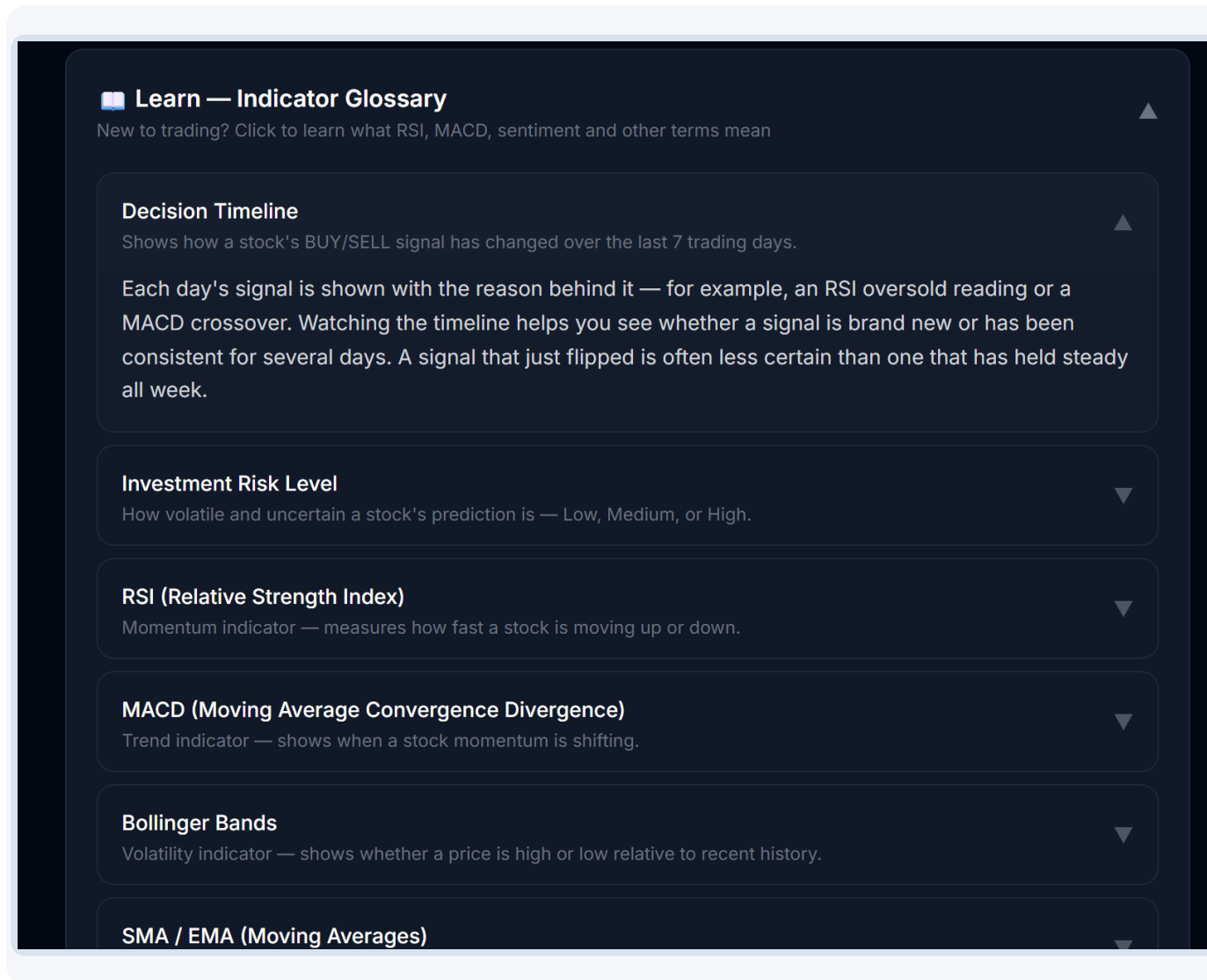
HOW TO READ IT

This is a history of model decisions, not a record of real trades. Its confidence labels are historical P(UP), including on SELL decisions.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/copilot_engine.py](#)

Learn tab: the indicator glossary



WHAT IT IS

The glossary helps a new user understand the terms used across Stock Detail.

WHERE DATA COMES FROM

Definitions are written in the frontend. This feature does not request a price forecast or use a language-model call.

HOW THE RESULT IS MADE

Open the glossary and select a term to reveal its plain-language explanation. The example expands the decision timeline.

HOW TO READ IT

Some older definitions say tomorrow or describe confidence loosely. The actual system target is five sessions, and historical confidence is P(UP). Use the current method explanation in this guide.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Detail.jsx](#)

Finance development: compare with a baseline

Development only · 2022–2024

2025 candidate test remains unopened

Do the models beat a simple baseline?

Three fixed finance models, separate calibration years and matching observations. Accuracy is shown alongside probability quality and the tendency to predict UP.

Validation observations	Instruments / fixed inputs	Predictor + calibration fits
32,472	44 / 13	9 + 9

No tested predictor beat the training-prior baseline on mean Brier loss.

The fixed selection rule chose Training-prior baseline. This is evidence from reused development years, not a final test or a reason to replace the live model.

Evaluation period

2022–2024 · equal-year mean

Download study JSON

WHAT IT IS

This study asks whether three fixed finance models outperform a simple training-frequency forecast.

WHERE DATA COMES FROM

A frozen Yahoo price snapshot supplies 44 instruments and 13 fixed finance inputs. Validation uses 2022, 2023 and 2024: 32,472 ticker-days.

HOW THE RESULT IS MADE

Logistic regression, histogram boosting and XGBoost use chronological fit, separate calibration and validation periods. There are nine predictor fits and nine calibration fits.

HOW TO READ IT

These are development years. No tested predictor beat the training-prior baseline on equal-year mean Brier loss, so none was promoted to the live system.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: [backend/data/research_reports/development_models.json](#)

Finance development: probability quality

Evaluation period

2022–2024 · equal-year mean

Download study JSON

Candidate	Accuracy ↑	Balanced acc. ↑	MCC ↑	ROC-AUC ↑	Brier loss ↓
Training-prior baseline	52.38%	50.00%	0.000	0.500	0.250877
Recent-prior baseline	48.90%	50.00%	0.000	0.500	0.253134
Always UP (P=100%)	52.38%	50.00%	0.000	0.500	0.476156
Logistic · raw	52.53%	50.19%	0.017	0.498	0.251299
Logistic · calibrated	49.06%	50.11%	0.015	0.493	0.253661
Histogram boosting · raw	52.50%	50.28%	0.011	0.502	0.252198
Histogram boosting · calibrated	48.84%	49.91%	-0.015	0.493	0.253701
XGBoost · raw	52.54%	50.27%	0.011	0.506	0.251999
XGBoost · calibrated	48.90%	49.89%	-0.012	0.497	0.254025

Lower Brier loss means better probability forecasts overall, not calibration alone. Balanced accuracy weights UP and DOWN equally; MCC measures agreement beyond chance (–1 to 1). The training-prior baseline gives every instrument the UP frequency from the fit period; the recent-prior baseline uses the calibration year. Always UP uses P(UP)=100%, so its probability loss can be poor even when its directional accuracy is reasonable.

WHAT IT IS

The table compares direction accuracy and the quality of the probability forecasts on matching observations.

WHERE DATA COMES FROM

The study JSON contains raw and calibrated versions of the three predictors plus training-prior, recent-prior and always-UP baselines.

HOW THE RESULT IS MADE

Choose an individual validation year or the equal-year mean. Brier is the mean squared probability error; smaller is better. Other columns show balance, MCC, AUC and UP rate.

HOW TO READ IT

The training-prior baseline has mean Brier 0.250877, versus 0.251999 for raw XGBoost. A similar accuracy score does not imply a better probability forecast.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: [backend/data/research_reports/development_models.json](#)

Confidence audit: accuracy versus coverage

Minimum confidence	Retained rows	Coverage	Signal dates	Mean confidence	Accuracy	Baseline
≥ 50.00%	32,472	100.00%	738	56.29%	52.53%	52.53%
≥ 55.00%	20,449	62.97%	725	58.17%	52.20%	52.20%
≥ 60.00%	4,193	12.91%	389	62.22%	51.51%	51.51%
≥ 65.00%	375	1.15%	94	66.93%	56.80%	56.80%
≥ 70.00%	32 Small sample	0.10%	24	73.64%	62.50%	62.50%
≥ 80.00%	0	0.00%	0	—	—	—
≥ 90.00%	0	0.00%	0	—	—	—

All thresholds are fixed diagnostics. No threshold is recommended or applied to live signals. Small sample marks fewer than 200 rows or 20 signal dates; a larger sample still does not establish reliability. Empty selections show undefined scores.

Pooled scores weight each retained observation, unlike the equal-year means above. Samples differ across candidates and thresholds, so compare each with its matching baseline. These previously studied years, overlapping outcomes and dependent stocks do not provide a fresh test or significance guarantee.

WHAT IT IS

This diagnostic asks what is lost or gained when lower-confidence predictions are removed.

WHERE DATA COMES FROM

It reuses the fixed development predictions. It compares nine outputs, seven confidence thresholds and four period views, without fitting new models.

HOW THE RESULT IS MADE

Confidence is max(P(UP), P(DOWN)). For every retained set, the training-prior baseline is evaluated on exactly the same rows. Coverage is retained rows divided by eligible rows.

HOW TO READ IT

At the raw XGBoost 60% threshold, only 12.91% of rows remain and accuracy is 51.51%. Higher confidence did not guarantee better results. No cutoff is recommended or applied live.

Controlled news study

Exploratory study

Five-session direction • 2023 evaluation

Does financial news improve prediction?

Finance, news coverage, VADER and FinBERT compared on the same observations. The constant-probability baseline stays visible so model complexity has to earn its place.

Evaluation observations

10,780

Observations with news

3,658

Instruments covered by news

26 / 44

Prediction model

Logistic regression

Inputs	Accuracy	Balanced accuracy	Brier loss ↓
Constant training prior	55.23%	50.00%	0.247623
Finance	47.43%	51.41%	0.255580
Finance + coverage	47.43%	51.37%	0.255928
Finance + VADER	47.49%	51.41%	0.255890
Finance + FinBERT	47.33%	51.24%	0.256194

The constant baseline uses the training-period UP frequency. Always predicting UP scores 55.23% accuracy. Balanced accuracy weighs UP and DOWN equally.

WHAT IT IS

This separate experiment compares finance, news coverage, VADER and FinBERT on the same historical observations.

WHERE DATA COMES FROM

A timestamped financial-news archive supplies historical text. The 2023 comparison has 10,780 evaluated ticker-days, 3,658 with news and coverage for 26 of 44 instruments.

HOW THE RESULT IS MADE

The study adds coverage and sentiment inputs under a fixed time-availability rule. The model selector switches between logistic regression and histogram boosting.

HOW TO READ IT

The screenshot uses logistic regression. Its constant training-prior baseline is stronger on Brier than the tested news variants. This study is exploratory and does not update serving models.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: docs/research/FINBERT_STUDY.md

FinBERT versus VADER: inspect uncertainty

FinBERT versus VADER

No clear improvement over VADER: the interval includes zero.

Brier improvement: -0.000304 · 95% block-bootstrap interval: [-0.000706, 0.000046]. Positive values favor FinBERT.
These intervals are exploratory and do not correct for multiple comparisons.

► Method and limitations

WHAT IT IS

This panel checks whether the finance-language sentiment model gives a clear improvement over the simpler VADER scorer.

WHERE DATA COMES FROM

Both variants use the same research rows and news timing. The report stores their Brier difference and a block-bootstrap interval.

HOW THE RESULT IS MADE

A positive Brier improvement favours FinBERT. Blocks preserve some time dependence when resampling the observed period.

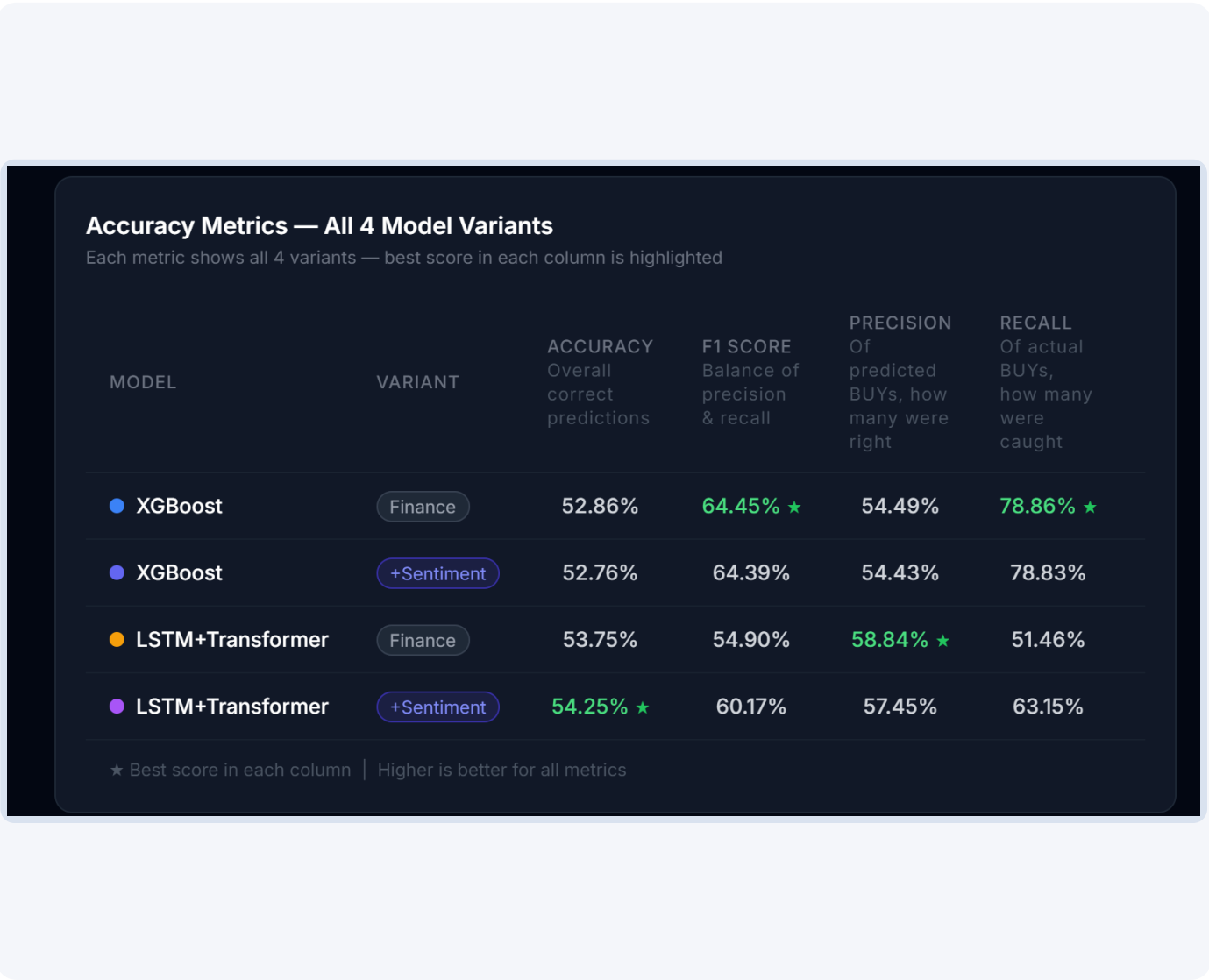
HOW TO READ IT

For the displayed logistic comparison, the interval includes zero, so there is no clear improvement. Results depend on the selected predictor and do not prove better live accuracy.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/components/ResearchComparison.jsx](#)

Historical core models: table and bar chart



WHAT IT IS

The historical comparison displays accuracy, F1, precision, recall and AUC for the four core variants.

WHERE DATA COMES FROM

Saved model_metrics.json supplies these figures. The adjacent bar chart plots the same measurements side by side.

HOW THE RESULT IS MADE

A star marks the largest saved score in each column. The LSTM +Sentiment variant has 54.25% saved accuracy; finance-only LSTM has 53.75%.

HOW TO READ IT

These are older reused-period results, not the controlled development study above. Legacy LSTM+Transformer labels refer to LSTM with attention; the separate Transformer Encoder is another model.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: [backend/data/predictions/model_metrics.json](#)

Historical profiles: bars, differences and radar

WHAT IT IS

The charts make trade-offs between the four saved core variants easier to see.

WHERE DATA COMES FROM

All bars, variant differences and radar axes are derived from the same historical metrics file.

HOW THE RESULT IS MADE

Bars compare each metric directly. Variant differences subtract finance-only scores from text-feature scores. The radar view plots all five metrics as a profile.

HOW TO READ IT

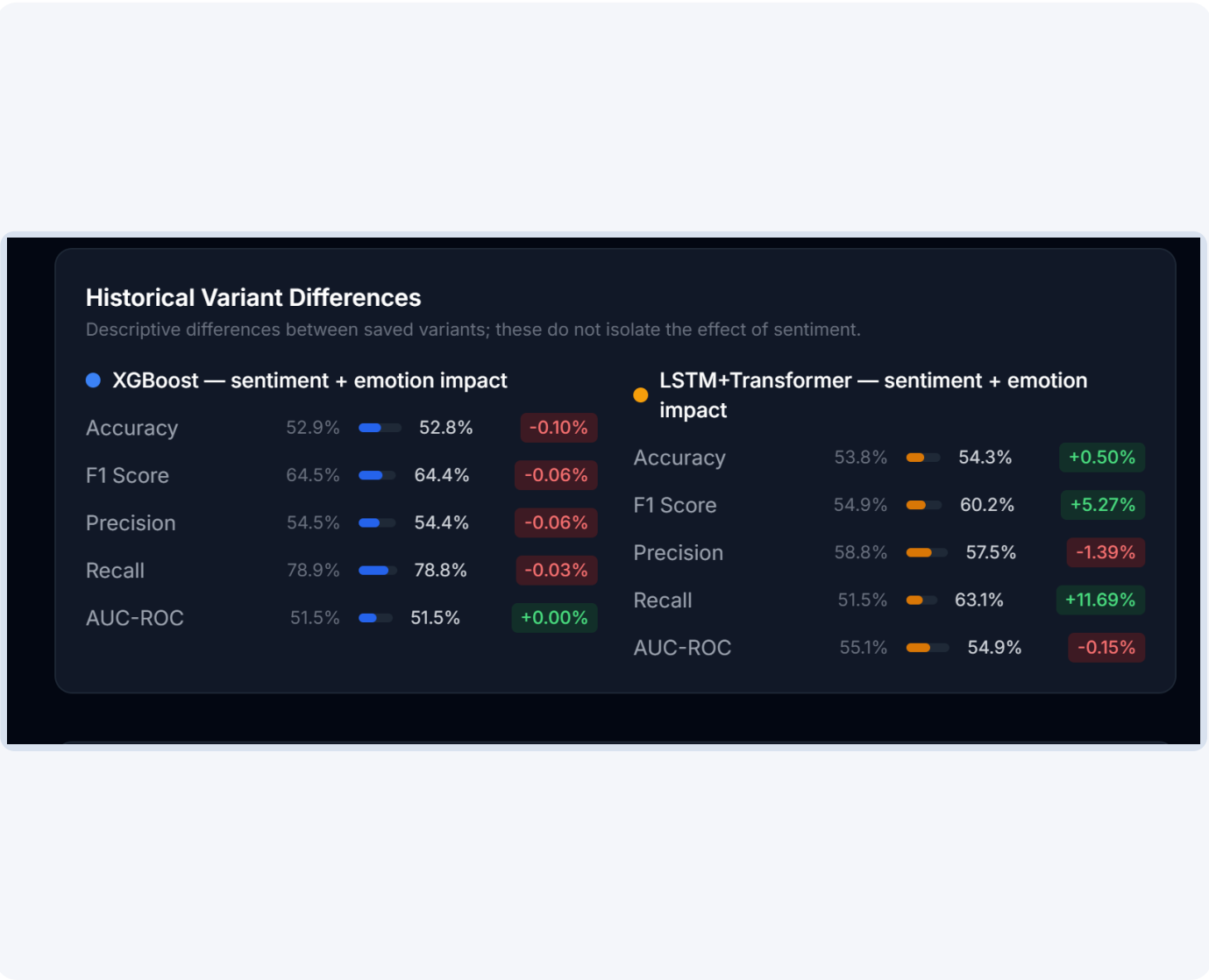
Chart size is not evidence of robustness. A positive variant difference cannot isolate the effect of sentiment when the evaluated historical text inputs are zero.



REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Compare.jsx](#)

Historical variant differences



WHAT IT IS

This section shows the numerical differences between the saved finance-only and text-feature variants.

WHERE DATA COMES FROM

The interface reads the historical metric values for XGBoost and LSTM, then compares each pair.

HOW THE RESULT IS MADE

Each difference is the text-feature score minus its finance-only counterpart. For accuracy, a difference of 0.50 means 0.50 percentage points.

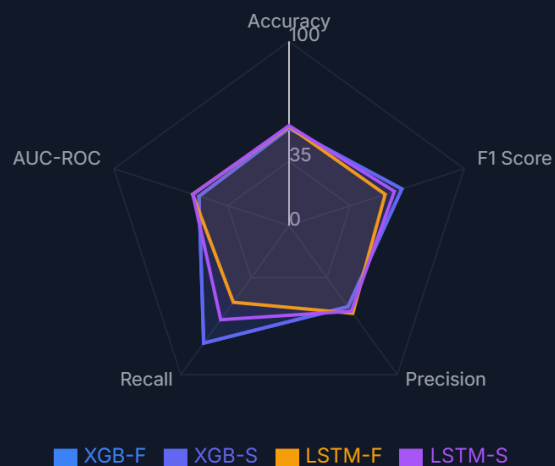
HOW TO READ IT

The page does not establish that adding news caused the difference. Different training behaviour and zero text coverage can still produce different model outputs.

Radar chart: model profiles

Model Profile — Radar Chart

Spider chart view — shows relative strengths of each model across all metrics



WHAT IT IS

The radar chart compares the four core models' historical metric profiles.

WHERE DATA COMES FROM

Its axes use the same saved accuracy, F1, precision, recall and AUC values as the metrics table.

HOW THE RESULT IS MADE

Each coloured shape plots one model's scores on five axes. The legend identifies the finance-only and text-feature variants.

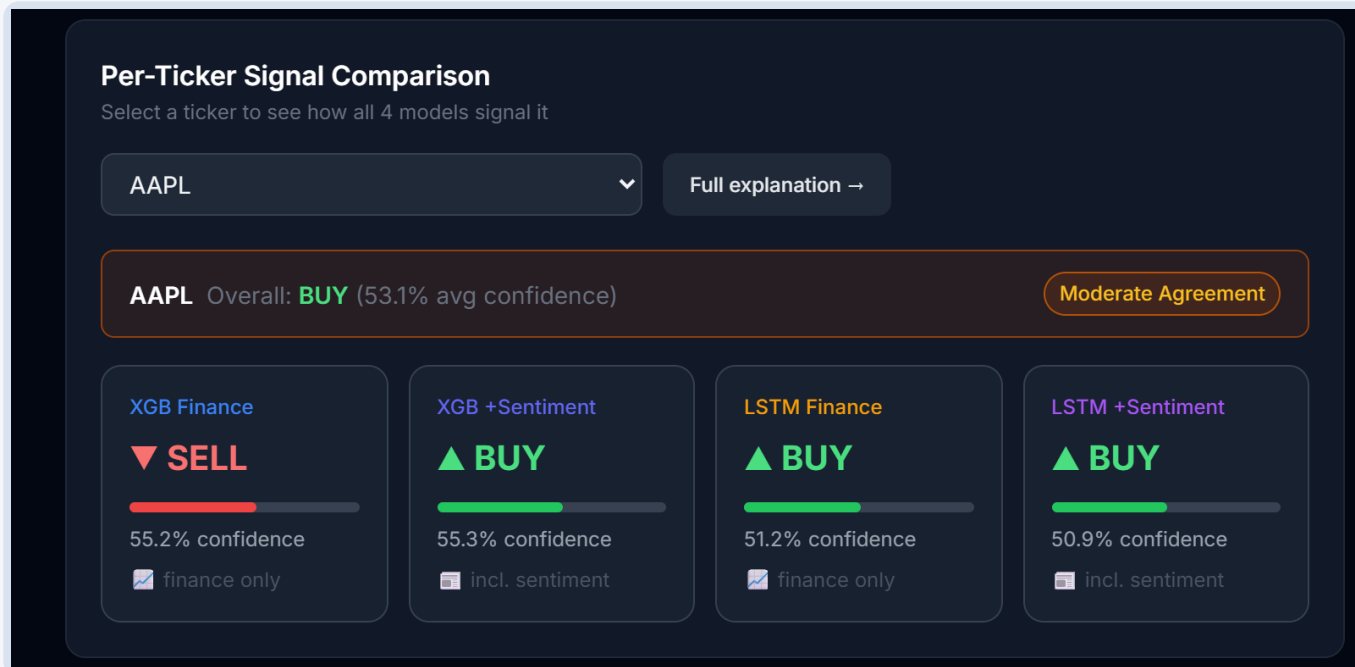
HOW TO READ IT

A bigger shape is a visual summary, not a statistical test. Metrics have different meanings, and the chart does not establish better future portfolio returns.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Compare.jsx](#)

Per-ticker model comparison



WHAT IT IS

The ticker selector compares the four core models' saved decisions for one instrument.

WHERE DATA COMES FROM

Selecting AAPL calls the ticker-comparison API for the saved model signals, probabilities and variant differences.

HOW THE RESULT IS MADE

Each model card shows its BUY/SELL label and historical P(UP). The comparison lets the user inspect disagreement hidden by an overall vote.

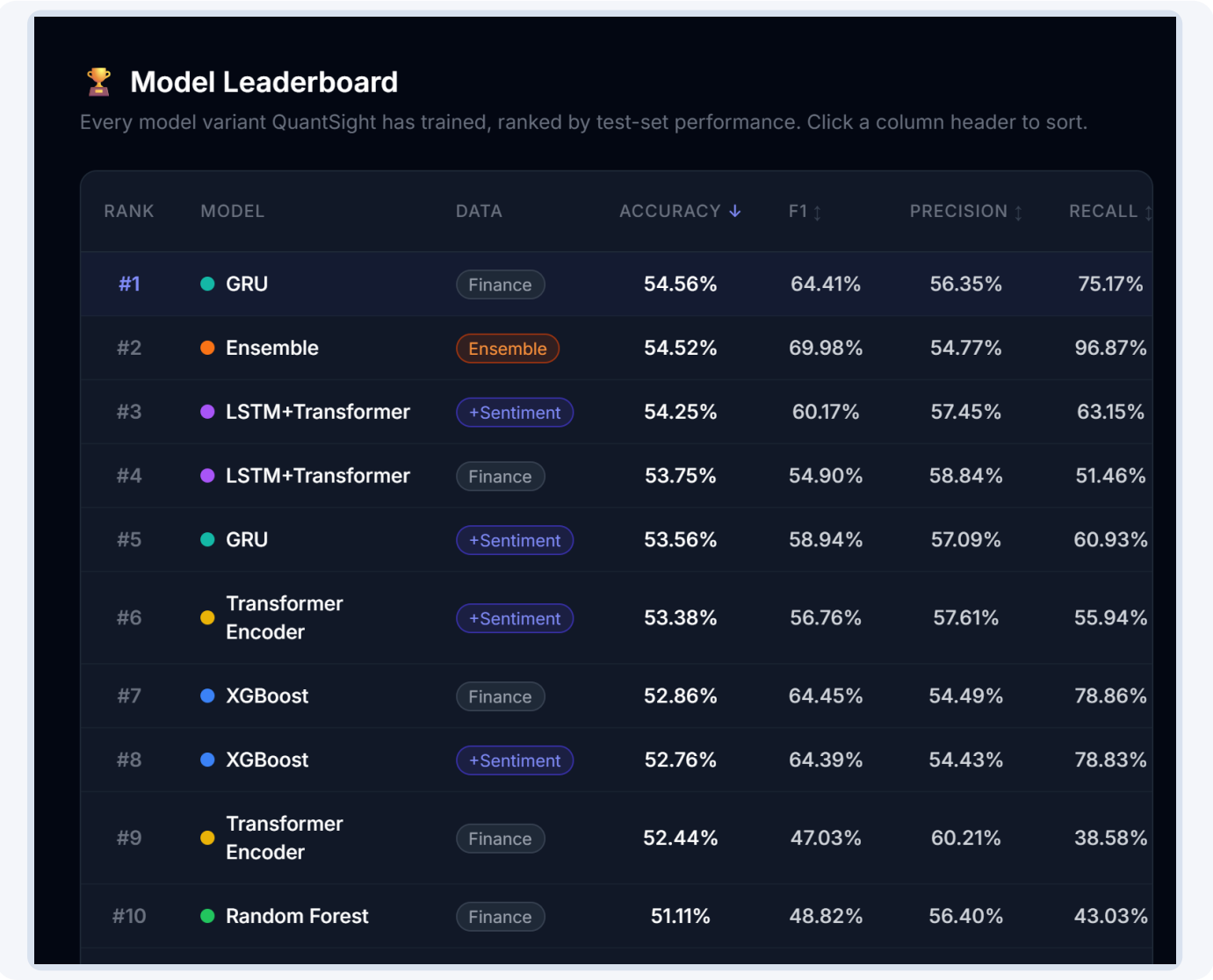
HOW TO READ IT

This uses saved historical outputs. It does not retrain the models or produce four current live predictions. A difference between text and finance variants does not prove news caused a gain.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Compare.jsx](#)

Model leaderboard: all trained variants



WHAT IT IS

The leaderboard ranks six model families, their two input variants and the four-core-model probability ensemble: 13 entries.

WHERE DATA COMES FROM

The metrics API reads the stored model results. Column headers allow sorting by accuracy, F1, precision, recall or AUC.

HOW THE RESULT IS MADE

Sorting changes which model leads for the chosen metric. GRU Finance has the highest saved classification accuracy, 54.56%, in this file.

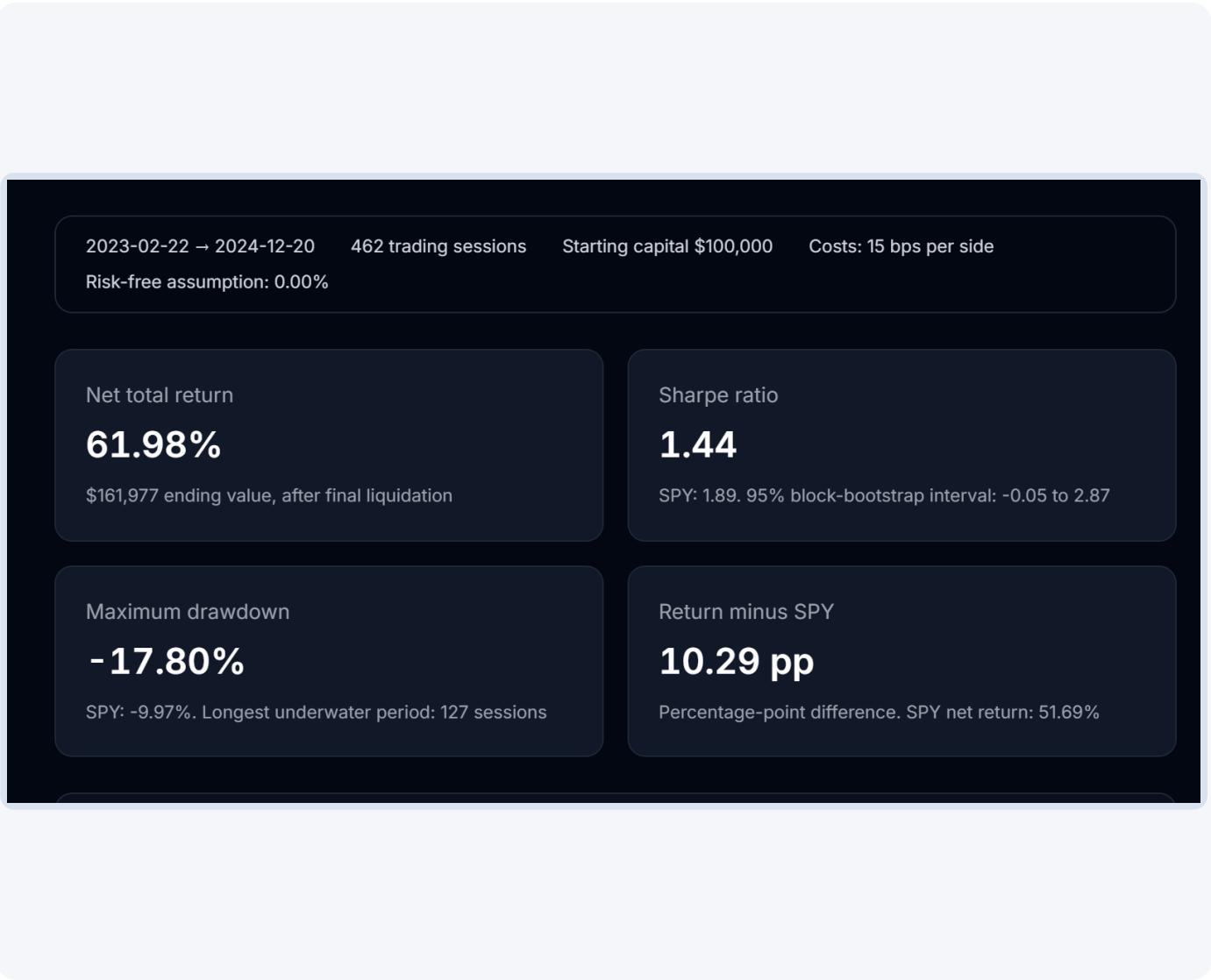
HOW TO READ IT

This is a model leaderboard, separate from player scores. Its static 50% reference is only a visual guide; always-UP and training-prior baselines are more informative for an imbalanced target.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Leaderboard.jsx](#)

Backtesting: settings and net performance



WHAT IT IS

Backtesting simulates a clearly specified trading policy using the saved model predictions.

WHERE DATA COMES FROM

The report combines saved predictions and historical Open/Close prices for 44 instruments, from 22 February 2023 to 20 December 2024: 462 sessions.

HOW THE RESULT IS MADE

Starting capital is \$100,000. Default costs are 10 bps commission plus 5 bps slippage per side. Selecting a strategy loads its saved results and matching benchmarks.

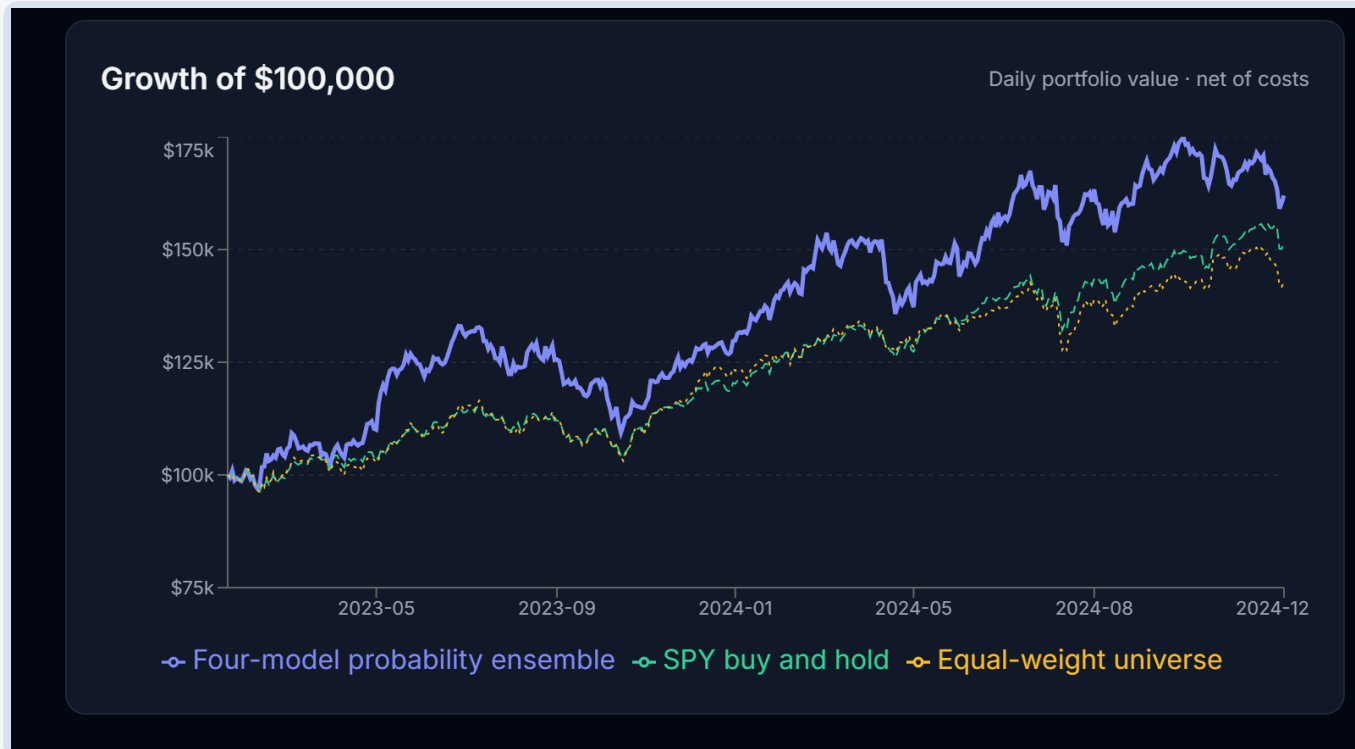
HOW TO READ IT

The default ensemble shows 61.98% net return, Sharpe 1.44 and maximum drawdown -17.80%. These are exploratory simulations after costs, not a verified live trading advantage.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/data/backtests/report.json](#)

Backtesting: growth of capital



WHAT IT IS

The equity chart shows what the simulated account was worth at each daily close.

WHERE DATA COMES FROM

Daily values are saved by the backtest engine. Selected strategy, SPY buy-and-hold and the equal-weight universe use the same observed dates.

HOW THE RESULT IS MADE

Equity equals cash plus marked holding value. Shares remain invested between rebalances, so opening gaps and weekend changes are included. Costs reduce cash.

HOW TO READ IT

A higher final value can come with larger losses along the way. Charts show historical simulation; they do not represent money invested by the project.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/backtesting/engine.py](#)

Backtesting: drawdown and downside risk

Decline from previous peak



Annualized return (CAGR)

30.09%

Annualized volatility: 19.55%

Sortino / Calmar

2.15 / 1.69

Return relative to downside variation / worst drawdown

Average market exposure

100.00%

Costs paid: \$19,132 · 708 executions

Positive-return days

54.76%

Daily portfolio hit rate, not the percentage of winning trades

WHAT IT IS

Drawdown measures how far the account fell below its previous peak.

WHERE DATA COMES FROM

The engine derives drawdown, volatility, Sortino and Calmar from the continuous daily equity and net-return series.

HOW THE RESULT IS MADE

Drawdown is $\text{equity} / \text{previous peak} - 1$. The ensemble's worst decline is 17.80%, with a longest underwater period of 127 sessions. Initial capital is included as a peak.

HOW TO READ IT

Sharpe and downside ratios depend on the period and the explicit 0% risk-free assumption. The ensemble has more total return than SPY here but a lower Sharpe and worse drawdown.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/backtesting/engine.py](#)

Backtesting: year-by-year consistency

Performance over time

Four-model probability ensemble: check whether the overall result holds across different periods.

Returns cover the dates shown, including partial years and months. They include trading costs and are not annualized. Holdings continue across period boundaries.

Months beating SPY
9 / 23
0 tied; descriptive consistency, not prediction accuracy.

Best observed month
2023-05: 12.23%

Worst observed month
2024-04: -8.93%

Year breakdown · observed study windows

PERIOD	OBSERVED DATES	SESSIONS	SELECTED NET RETURN	SPY	EQUAL-WEIGHT UNIVERSE	RETURN MINUS SPY
2023	2023-02-22 to 2023-12-29	216	29.40%	20.60%	23.41%	+8.80 pp
2024	2024-01-02 to 2024-12-20	246	25.18%	25.78%	15.78%	-0.60 pp

► Monthly comparison · 23 observed months

WHAT IT IS

Period performance checks whether the overall result holds across different parts of the study.

WHERE DATA COMES FROM

The period report compounds the same saved daily net returns. Holdings and costs continue across year and month boundaries.

HOW THE RESULT IS MADE

In the observed 2023 window, ensemble return is 29.40% versus SPY 20.60%. In 2024 through 20 December, it is 25.18% versus SPY 25.78%.

HOW TO READ IT

These are partial observed years, not annualised returns. The ensemble beat SPY in 9 of 23 months; a strong total return does not mean it won consistently.

Backtesting: monthly breakdown

▼ Monthly comparison · 23 observed months

PERIOD	OBSERVED DATES	SESSIONS	SELECTED NET RETURN	SPY	EQUAL- WEIGHT UNIVERSE	RE MI SP
2023-02	2023-02-22 to 2023-02-28	5	-1.09%	-0.96%	-1.07%	-0
2023-03	2023-03-01 to 2023-03-31	23	10.46%	3.71%	4.02%	+6
2023-04	2023-04-03 to 2023-04-28	19	-3.06%	1.60%	0.03%	-4
2023-05	2023-05-01 to 2023-05-31	22	12.23%	0.46%	1.75%	+1
2023-06	2023-06-01 to 2023-06-30	21	6.01%	6.48%	5.58%	-0
2023-07	2023-07-03 to 2023-07-31	20	5.37%	3.27%	5.33%	+2
2023-08	2023-08-01 to 2023-08-31	23	-2.96%	-1.63%	-3.38%	-1
2023-09	2023-09-01 to 2023-09-29	20	-7.34%	-4.74%	-3.60%	-2
2023-10	2023-10-02 to 2023-10-31	22	-5.36%	-2.17%	-2.83%	-3
2023-11	2023-11-01 to 2023-11-30	21	10.43%	9.13%	10.10%	+1
2023-12	2023-12-01 to 2023-12-29	20	3.70%	4.57%	6.34%	-0

WHAT IT IS

The expanded monthly table helps identify good and weak periods instead of relying only on one final score.

WHERE DATA COMES FROM

Monthly returns come from the period report built from continuous daily equity. The table contains 23 observed months; this screenshot shows its first section.

HOW THE RESULT IS MADE

The net return is the product of $(1 + \text{daily return})$ minus 1. The best ensemble month is May 2023, +12.23%; the worst is April 2024, -8.93%.

HOW TO READ IT

Month returns compound; they should not be added. Partial first and last months must be read using their observed dates. This is descriptive evidence, not a significance test.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/backtesting/periods.py](#)

Backtesting: compare all strategies

All strategies · same evaluation period

Select a strategy to update the charts. Scroll horizontally to see all metrics.

STRATEGY	NET RETURN	CAGR	SHARPE	SORTINO	MAX DRAWDOWN	EXPOSURE
SPY buy and hold	51.69%	25.52%	1.89	2.82	-9.97%	100.00%
Equal-weight universe (weekly)	42.89%	21.49%	1.52	2.27	-11.48%	100.00%
XGBoost - Finance	33.83%	17.23%	1.09	1.62	-14.51%	100.00%
XGBoost + Sentiment	45.44%	22.67%	1.35	2.08	-12.92%	100.00%
LSTM - Finance	72.16%	34.49%	1.74	2.59	-9.69%	100.00%
LSTM + Sentiment	67.59%	32.53%	1.60	2.37	-12.18%	100.00%
GRU - Finance	113.51%	51.24%	2.45	3.91	-8.96%	96.58%
GRU + Sentiment	73.27%	34.96%	1.56	2.56	-18.13%	90.65%
Random Forest - Finance	26.06%	13.47%	0.85	1.26	-14.12%	100.00%

WHAT IT IS

The strategy table compares 12 individual model variants, the core-model ensemble and two benchmarks.

WHERE DATA COMES FROM

All 15 portfolios come from the same saved run, with matching evaluation dates and transaction-cost accounting.

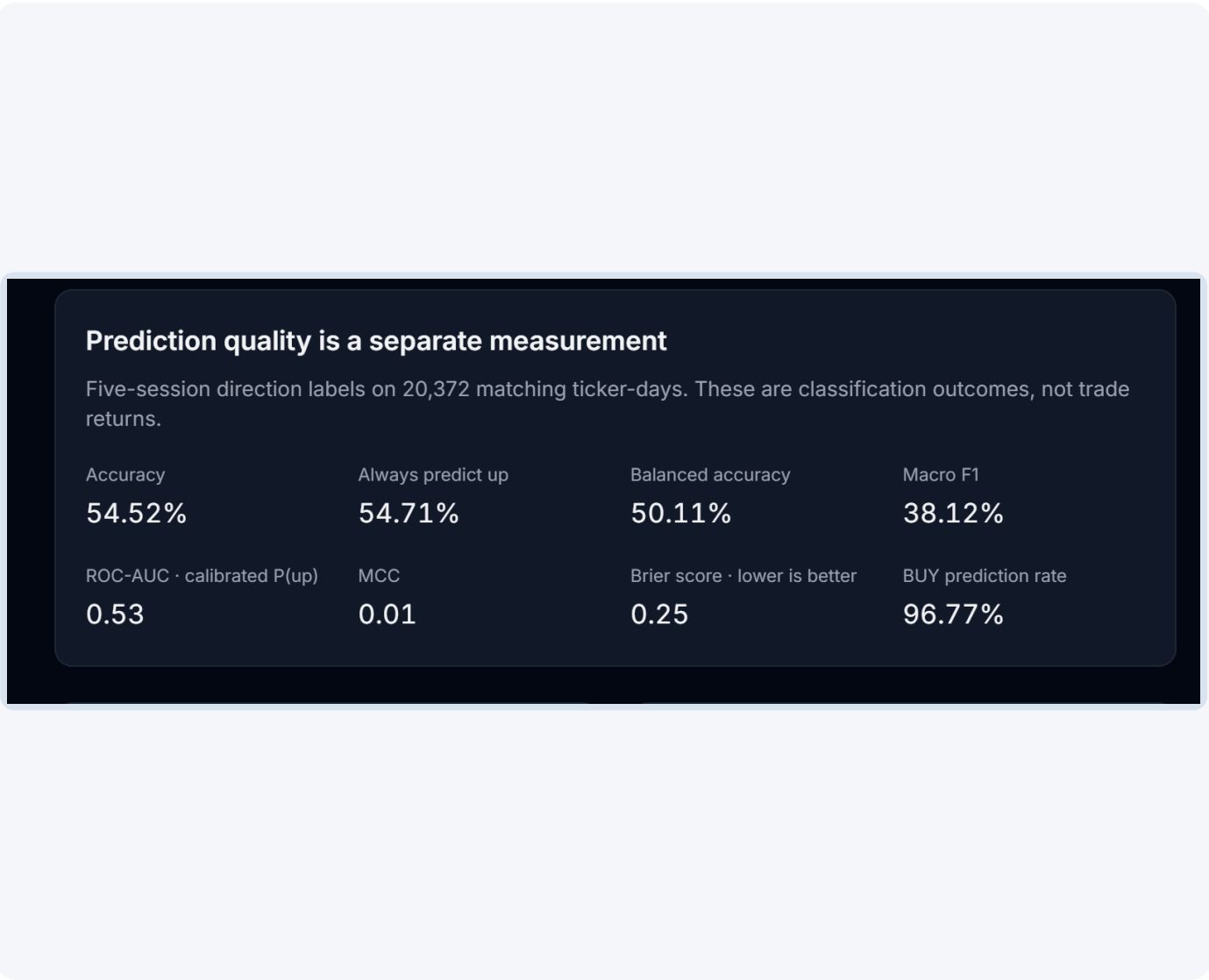
HOW THE RESULT IS MADE

Columns show net return, CAGR, Sharpe, Sortino, maximum drawdown and exposure. Selecting a strategy changes its charts. More metrics can be viewed by scrolling.

HOW TO READ IT

GRU Finance has the strongest historical return in this run, but selecting it after seeing results would introduce selection bias. Strategy rankings do not prove future performance.

Backtesting: classification is separate



WHAT IT IS

This card measures the models' five-session direction predictions, rather than the simulated portfolio's returns.

WHERE DATA COMES FROM

The engine aligns 20,372 common ticker-days and compares model decisions with five-session outcome labels.

HOW THE RESULT IS MADE

The ensemble has 54.52% accuracy, while always predicting UP has 54.71% on the same rows. Balanced accuracy, macro F1, AUC, MCC, Brier and BUY rate add context.

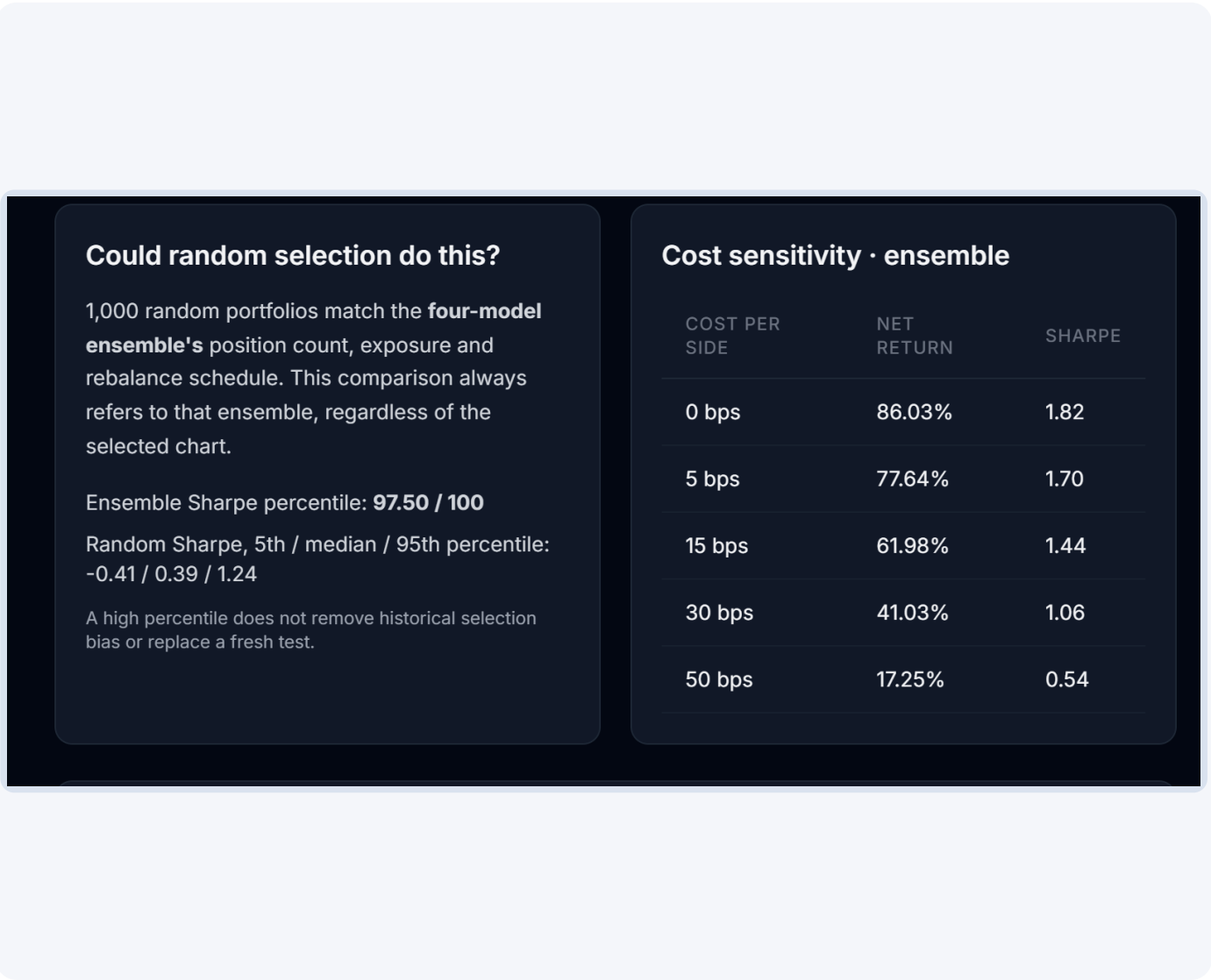
HOW TO READ IT

High BUY recall can result from predicting BUY almost all the time. Neither accuracy nor game score can be substituted for Sharpe or a net return.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/backtesting/engine.py](#)

Backtesting: random selection and trading costs



WHAT IT IS

These checks ask whether selection is better than a matched random policy and how sensitive returns are to costs.

WHERE DATA COMES FROM

The run includes 1,000 random portfolios matched to the ensemble's schedule, position count and exposure, plus fixed cost scenarios.

HOW THE RESULT IS MADE

The ensemble's Sharpe is at the 97.50th random percentile. Net return falls from 86.03% at zero cost to 17.25% at 50 bps per side.

HOW TO READ IT

Both panels always refer to the ensemble. The random test does not remove reused-history bias. Cost scenarios are assumptions, not measured broker spreads.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: [backend/data/backtests/report.json](#)

Backtesting: download the audit evidence

Download the evidence

Daily values and executions allow the portfolio calculations to be checked independently.

[All metrics CSV](#)[Selected daily values CSV](#)[Selected trades CSV](#)[Full report JSON](#)[Period report JSON](#)

Generated 2026-09-26 10:23:01 UTC · Run 20260926T102301904516Z

WHAT IT IS

Downloads make the simulation easier to reproduce and defend during an FYP discussion.

WHERE DATA COMES FROM

The published run stores metrics CSV, per-strategy daily values and trades CSV, the full report JSON and the period report JSON.

HOW THE RESULT IS MADE

Trade rows include signal and execution dates, ticker, quantity, price, traded amount, commission and slippage. Source hashes bind the report to its inputs.

HOW TO READ IT

Downloads refer to a saved run. The browser does not retrain models or silently rerun the simulation. Changed or missing sources are flagged instead of being mixed with old results.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: docs/BACKTESTING.md](#)

Live Track Record: waiting for real outcomes

Live Track Record

Predictions use a completed daily close and are recorded before the next market open. Each is checked against the close five NYSE trading sessions later. Historical backtests and old records with an unknown horizon are excluded from these results. Database administrators remain trusted.

—

Live Accuracy

five-session direction
accuracy

0

Resolved

checked against real
outcomes

44

Pending

waiting for target
session close

44

Total Logged

predictions dated in the
last 30 days

Same-sample always-UP baseline: N/A / Brier score: N/A (lower is better).

Direction confidence is the calibrated probability assigned to the predicted direction. It can be below 50% because the current model selects direction using its raw score.

Window starts 2026-09-04. Daily five-session outcomes overlap; predictions are not independent observations. This measures direction, not portfolio returns.

WHAT IT IS

The forward tracker records forecasts before the next market open and checks them after five NYSE sessions.

WHERE DATA COMES FROM

The API reads live_predictions_v2 in Supabase. The captured window contains 44 logged predictions: 44 pending, zero resolved.

HOW THE RESULT IS MADE

Once the exact target close is available, the outcome is UP if target close exceeds entry close; otherwise DOWN. Accuracy uses resolved records only, with a matching always-UP baseline and Brier.

HOW TO READ IT

There is no live accuracy result yet. The 28 September forecasts target 5 October's US close. Overlapping five-session outcomes are dependent. A transient connection failure recovered during capture.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/prediction_tracker.py](#)

Live Track Record: dated prediction records

RECENT PREDICTIONS						
TICKER	DATA SESSION / RECORDED (UTC)	TARGET SESSION	SIGNAL	DIRECTION CONFIDENCE	ACTUAL	RESULT
SNAP Snap Inc.	2026-09-28 2026-09-29 01:45:57	2026-10-05	▲ BUY	55.2%	—	Pending
AAPL Apple Inc.	2026-09-28 2026-09-29 01:45:45	2026-10-05	▼ SELL	44.8%	—	Pending
IWM iShares Russell 2000 ETF	2026-09-28 2026-09-29 01:45:51	2026-10-05	▲ BUY	55.2%	—	Pending
TGT Target Corp.	2026-09-28 2026-09-29 01:45:57	2026-10-05	▲ BUY	55.2%	—	Pending
GS Goldman Sachs Group Inc.	2026-09-28 2026-09-29 01:45:51	2026-10-05	▼ SELL	44.9%	—	Pending

WHAT IT IS

The recent-prediction table makes each forecast and its timing visible.

WHERE DATA COMES FROM

Supabase stores the original data session, recording time, target session, direction and probabilities, along with model/input versions and source times.

HOW THE RESULT IS MADE

The server inserts a forecast only once for its ticker, session and protocol. It resolves against the exact fifth-session close when available. Legacy records with unknown horizons are excluded.

HOW TO READ IT

These rows are still pending. Frozen records support an audit, but database administrators remain trusted. Recording a forecast is not proof that a scheduled job or its later outcome resolution always succeeds.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/prediction_tracker.py](#)

Portfolio: add a holding

My Portfolio × Cancel

Track your holdings and get model-based recommendations

Add a stock to your portfolio

Ticker	Number of shares	Your buy price (USD)
AAPL	10	200

Add to Portfolio

WHAT IT IS

The tracker lets a user enter a ticker, number of shares and purchase price.

WHERE DATA COMES FROM

Inputs are entered by the user. Guest holdings stay in browser localStorage; signed-in holdings use the owner's Supabase portfolio_holdings rows.

HOW THE RESULT IS MADE

The form validates positive shares and price, then stores the holding. The example uses 10 AAPL shares at \$200 in an isolated guest session.

HOW TO READ IT

These inputs are a documentation example, not the owner's real portfolio. The project records holdings but does not place broker orders or verify a purchase.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Portfolio.jsx](#)

Portfolio: value, profit and model recommendation

My Portfolio

Track your holdings and get model-based recommendations

+ Add Stock

TOTAL VALUE

\$3336.90

1 of 1 priced

TOTAL RETURN

+66.85%

+\$1336.90 all-time

TODAY'S P&L

+\$29.03

from live-priced holdings only

Your Holdings

Click any ticker to see full AI analysis

▲ Current prices are fetched live from Yahoo Finance where available, shown with a ● live badge. If live data is unavailable, the last dataset price (Dec 2024) is used instead, shown as hist. Model signals are based on historical training data — not financial advice.

TICKER	SHARES	BUY PRICE	CURRENT PRICE	VALUE	P&L	MODEL SIGNAL
AAPL Apple Inc.	10	\$200.00	\$333.69 ● live +0.87% today	\$3336.90	+\$1336.90 (+66.84%)	▲ BUY (53.1%)

◆ HOLD — model says BUY, keep your position · ▲▲ BUY MORE — strong BUY signal + you're profitable · ▼ SELL — model says SELL, consider exiting

WHAT IT IS

Holdings show their current value, unrealised profit/loss and a rule-based recommendation.

WHERE DATA COMES FROM

Yahoo supplies a quote where available; otherwise the API marks the last saved historical price as a fallback. Model recommendations still use saved historical signals.

HOW THE RESULT IS MADE

Value = shares x quote. Profit/loss = shares x (quote - buy price). Return divides this profit/loss by the purchase cost. Recommendations use consensus, confidence and holding return.

HOW TO READ IT

Today's P&L currently estimates current value x reported daily percentage change; it is an approximation. Quote values can change, and combining current prices with old signals does not make the signals current.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: frontend/src/pages/Portfolio.jsx

Portfolio: suggested instruments

What to Consider Buying Next

⚠️ These are model signals only — not financial advice. Always do your own research before investing.

#1 DIA

Strong Agreement

▲ BUY

#2 TLT

TLT
iShares 20+ Year Treasury Bond ETF

Strong Agreement

▲ BUY

WHAT IT IS

The suggestion list highlights historical BUY signals outside the user's current holdings.

WHERE DATA COMES FROM

The explanation API supplies the same saved four-model consensus used elsewhere. Already-held tickers are excluded.

HOW THE RESULT IS MADE

The list prioritises instruments where all four core models agree on BUY. Users can open Stock Detail to inspect the evidence before drawing conclusions.

HOW TO READ IT

The list is based on old model outputs. It is not personalised investment advice, a portfolio optimiser or a verified list of stocks that will rise.

REAL LOCAL SCREENSHOT / GUEST SESSION

Code / evidence: frontend/src/pages/Portfolio.jsx

Prediction game: choose the challenge

DIFFICULTY

Easy

High-confidence signals — model is very sure

Medium

Mixed confidence — requires more intuition

Hard

Low-confidence signals — even the model is unsure

MODEL TO PLAY AGAINST

XGBoost Finance

XGBoost +Sentiment

LSTM Finance

LSTM +Sentiment

The game will show you a stock question and reveal whether the selected model predicted BUY or SELL after you answer.

WHAT IT IS

The game teaches users to interpret stock indicators by guessing a model's saved BUY/SELL decision.

WHERE DATA COMES FROM

Questions come from the saved core-model predictions and dated historical indicator rows. A separate public player leaderboard uses `game_progress`.

HOW THE RESULT IS MADE

Choose Easy, Medium or Hard and one of the four core models. Difficulty changes the confidence range used when sampling questions. B and S keyboard shortcuts also answer questions.

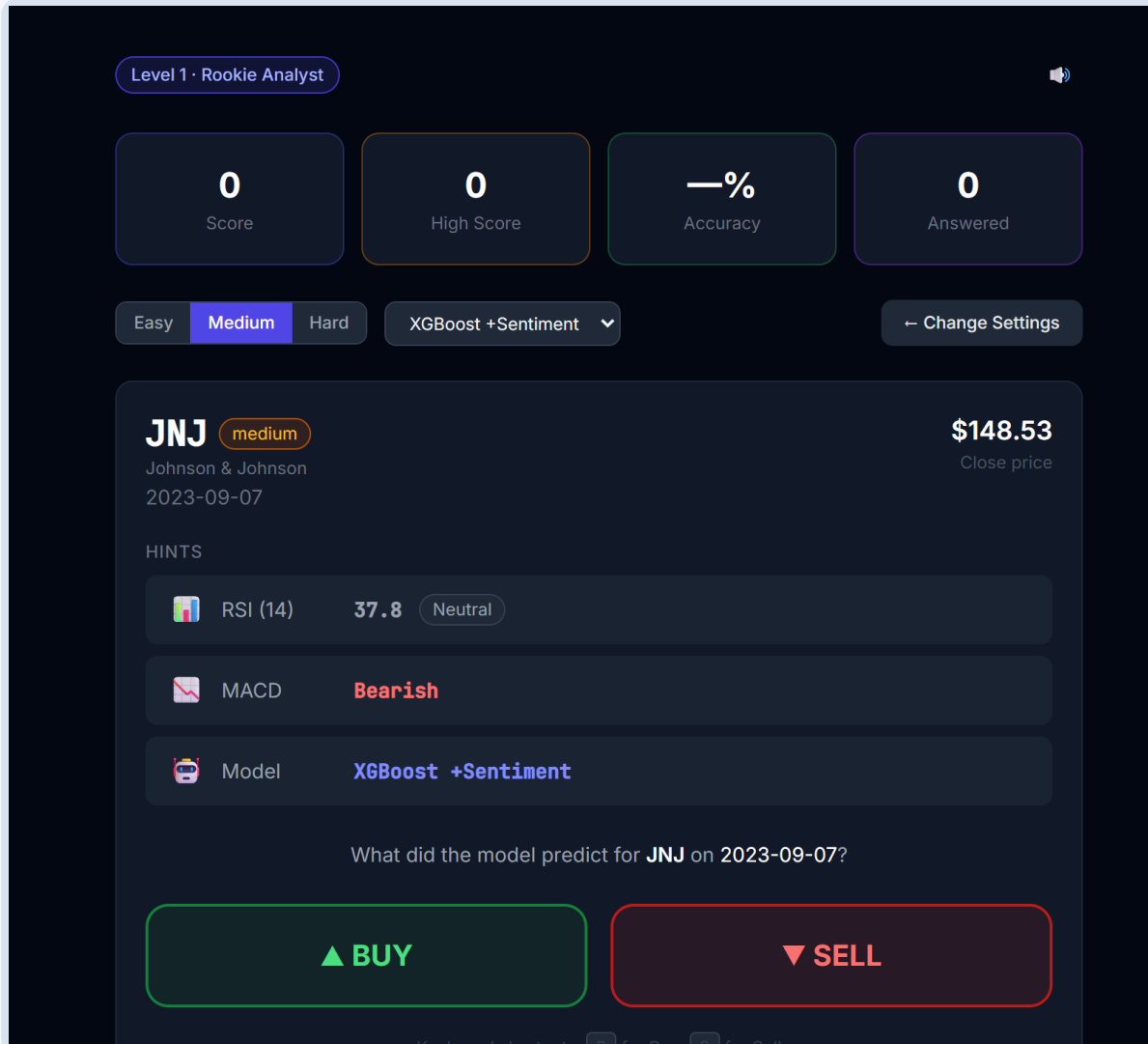
HOW TO READ IT

The game evaluates agreement with a model label, not the actual future market move. High game accuracy is not evidence that either the player or model predicts returns well.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code](#) / [evidence: frontend/src/pages/Game.jsx](#)

Prediction game: a real question



REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/main.py](#)

WHAT IT IS

A question shows a random ticker and historical date, a close price, RSI and MACD hints.

WHERE DATA COMES FROM

The question API chooses an eligible saved model row and reads the matching stock indicators for that date.

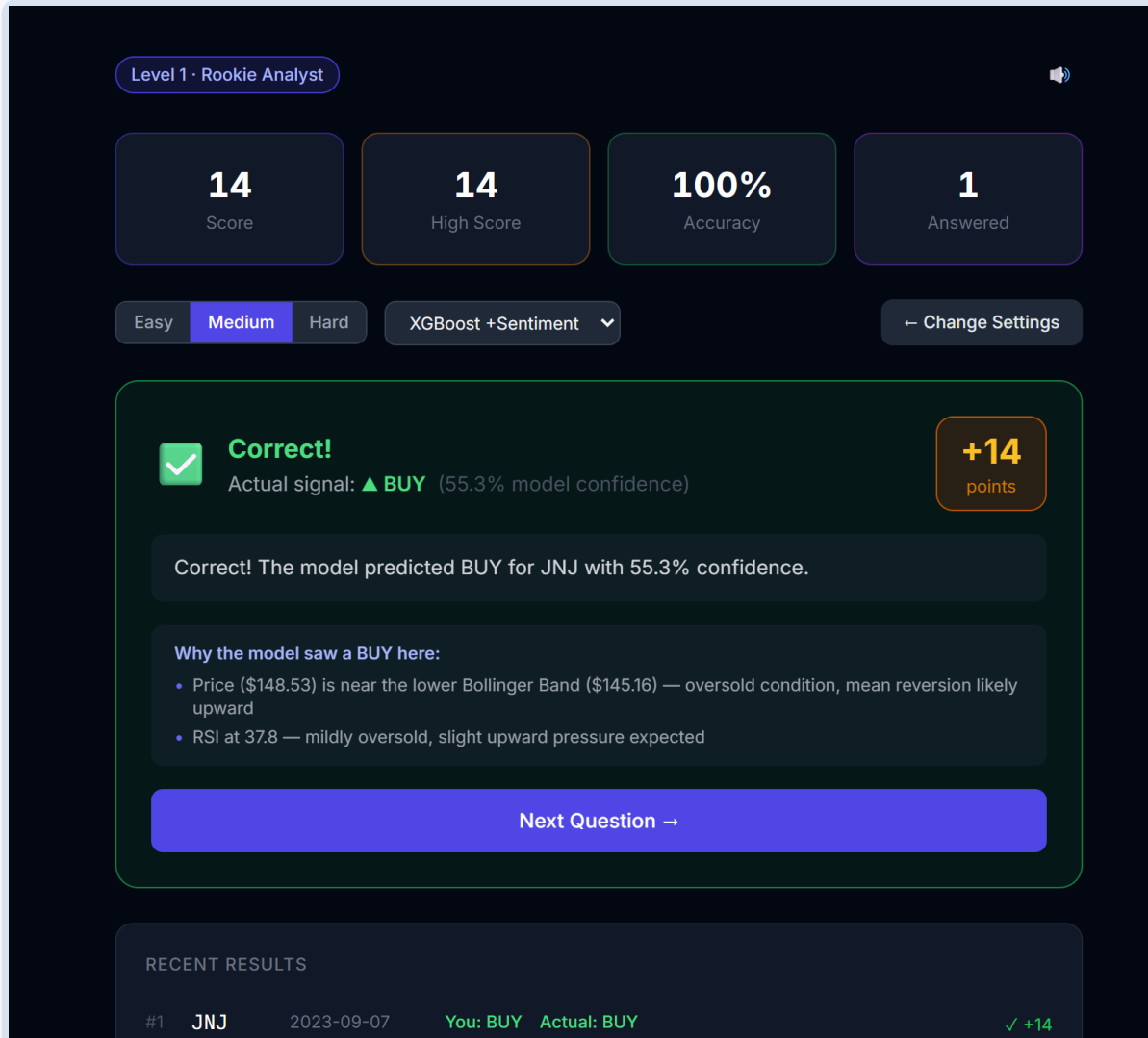
HOW THE RESULT IS MADE

The user selects BUY or SELL before the model label is revealed. The score, streak, level and answered count reflect guest or signed-in game progress.

HOW TO READ IT

This is an educational quiz. The displayed date is historical, and answering does not execute a trade or insert a live forecast.

Prediction game: result and explanation



WHAT IT IS

The result reveals whether the guess matched the selected model and offers short indicator-based reasons.

WHERE DATA COMES FROM

The answer API compares the guess with the saved model prediction. Guest progress is stored in the temporary browser; signed-in progress syncs to Supabase.

HOW THE RESULT IS MADE

A matching guess earns base points plus a difficulty bonus. Game accuracy is matching answers / answered questions. Recent results and high scores summarise play.

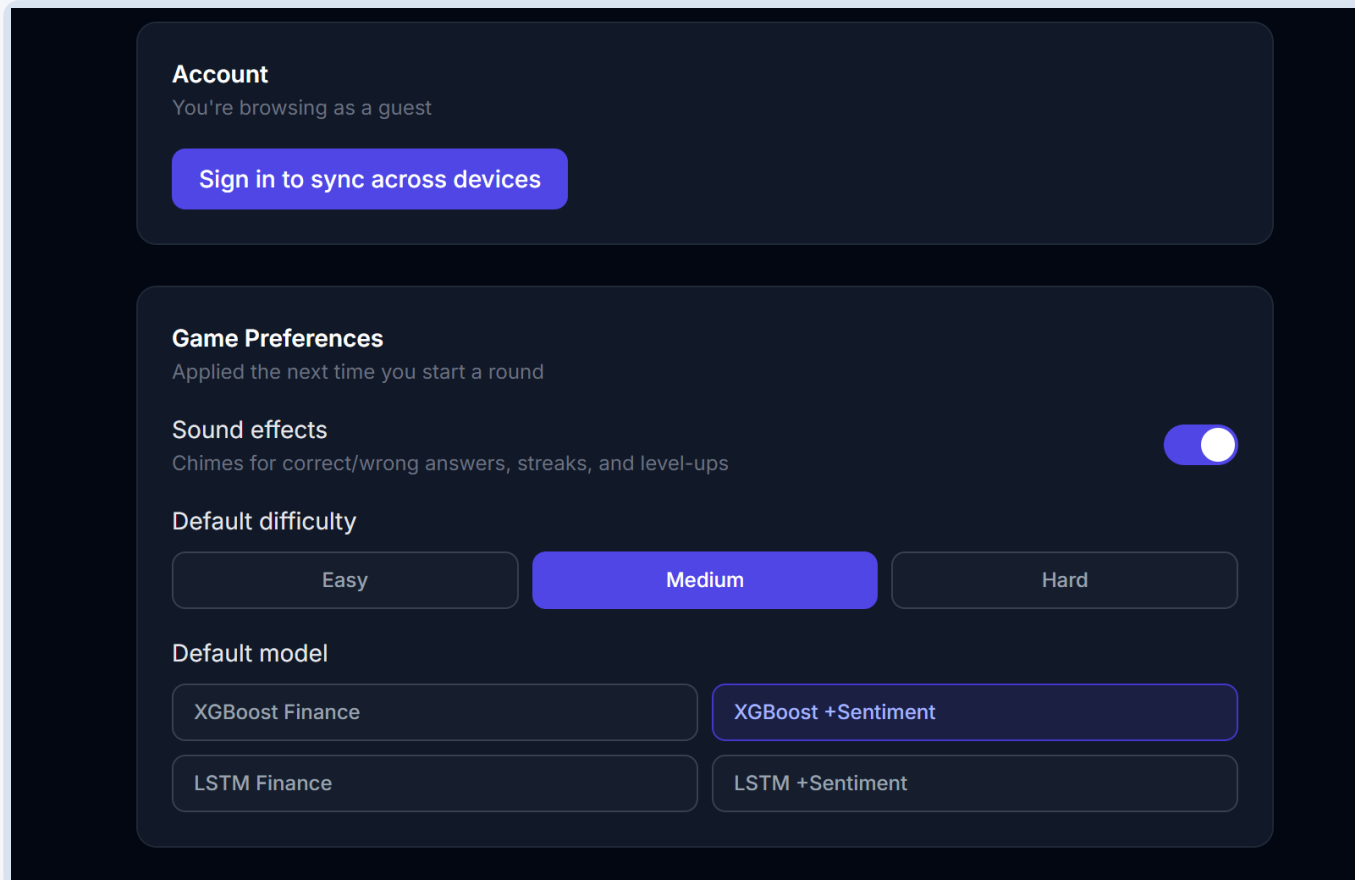
HOW TO READ IT

The interface's Actual signal means the model's label in this game, not the realised stock return. The shown guest round was answered for documentation only.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/main.py](#)

Settings: account and game preferences



WHAT IT IS

Settings lets users manage the account view and choose defaults for the next game round.

WHERE DATA COMES FROM

Game preferences are saved to this browser. Account identity, password and sign-out actions use Supabase Auth and profile fields.

HOW THE RESULT IS MADE

Controls set sound, default difficulty and default model. Signed-in account controls include display name, password change and sign-out options; the screenshot shows guest mode.

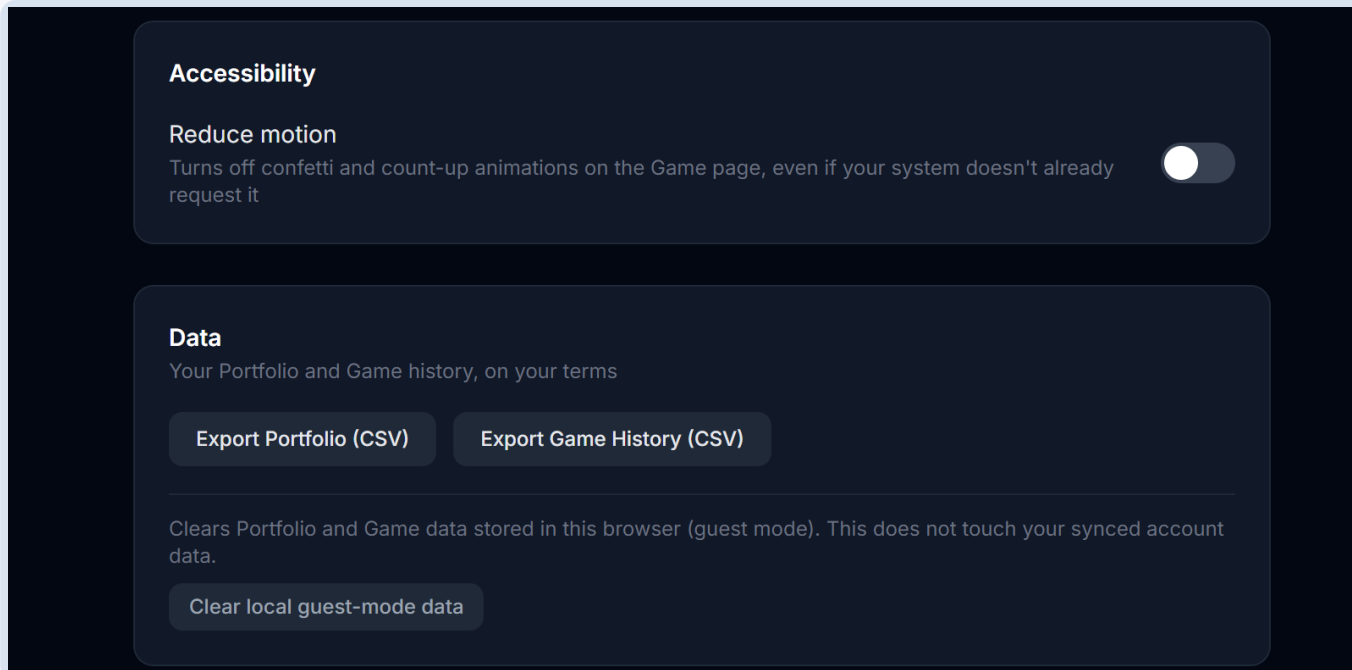
HOW TO READ IT

Authenticated account changes were reviewed in code, not submitted for this guide. They require a working Supabase session and the expected database fields.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Settings.jsx](#)

Settings: accessibility, exports and reset



WHAT IT IS

Reduce motion limits game animations. Exports help users keep a copy of their own holdings and game history.

WHERE DATA COMES FROM

The export buttons read the current guest data or signed-in account rows. The reset action targets guest Portfolio and Game keys in this browser.

HOW THE RESULT IS MADE

Export buttons create CSV files. Clear local guest-mode data uses a confirmation step and removes local guest records; synced account records are separate.

HOW TO READ IT

Exports describe recorded app data, not verified broker activity. Local guest storage is browser-specific and can be lost when browser data is cleared.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Settings.jsx](#)

Optional sign-in and account creation

Sign In

Optional — sync your Portfolio and Game progress across devices. Everything still works without an account.

 Continue with Google

or

Email

Password

Sign In

Don't have an account? [Create one](#)

WHAT IT IS

Users can browse as guests or sign in to sync Portfolio and Game progress across devices.

WHERE DATA COMES FROM

Supabase Auth handles email/password and Google sign-in. The frontend uses the public project client configuration.

HOW THE RESULT IS MADE

The form signs in an existing user. Create one switches to account registration. After authentication, app data uses owner-scoped database access.

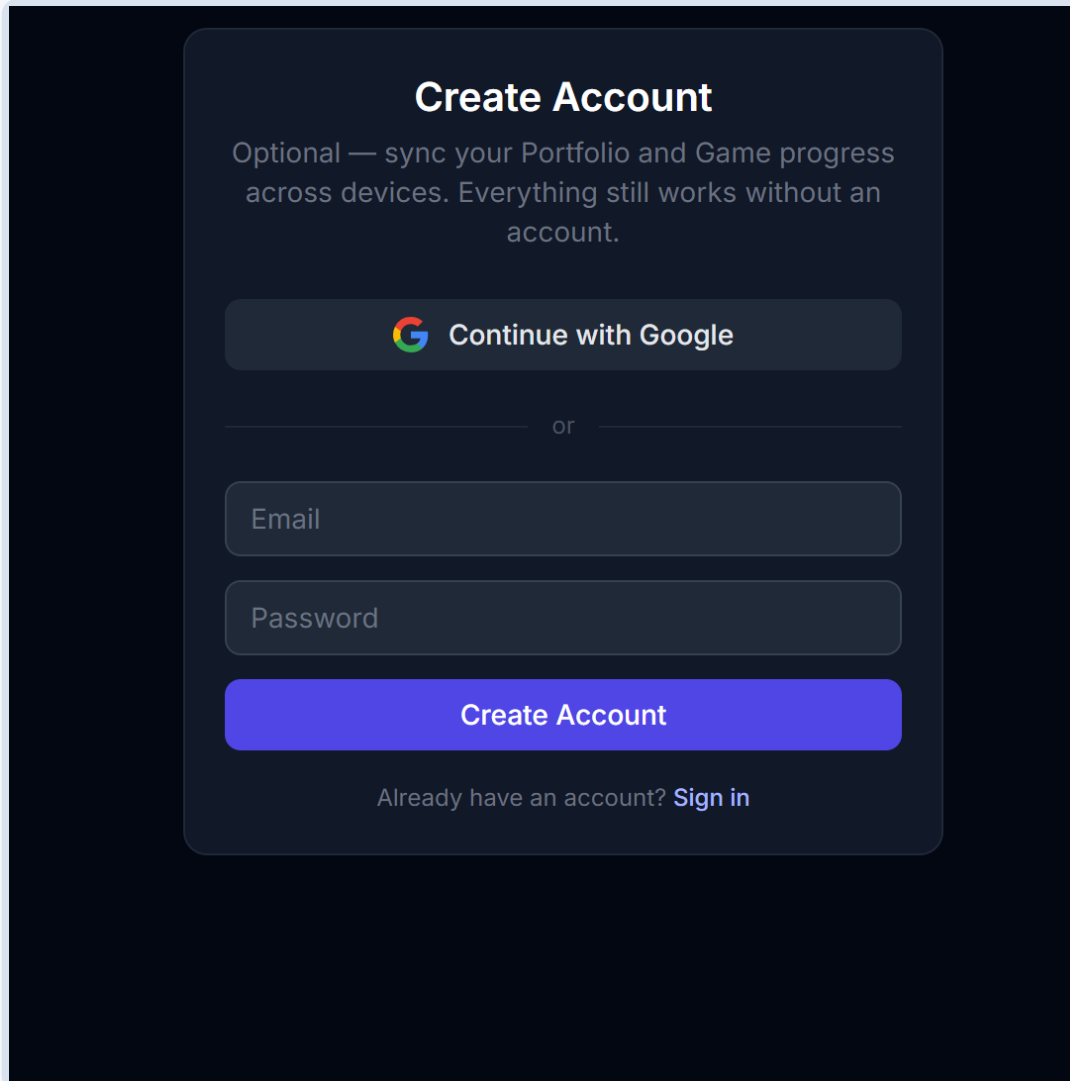
HOW TO READ IT

No account was created or signed into for these screenshots. Google redirects and email confirmation depend on the project's Supabase Auth configuration.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code](#) / [evidence: frontend/src/context/AuthContext.jsx](#)

Account registration form



The screenshot shows a dark-themed 'Create Account' form. At the top, the title 'Create Account' is in white. Below it, a subtitle reads: 'Optional — sync your Portfolio and Game progress across devices. Everything still works without an account.' The form features a 'Continue with Google' button with the Google logo, followed by a horizontal line with the word 'or' in the center. Below this are two input fields labeled 'Email' and 'Password'. A prominent blue 'Create Account' button is positioned below the input fields. At the bottom, a link says 'Already have an account? Sign in'.

WHAT IT IS

The registration view collects the details required to create an optional synced account.

WHERE DATA COMES FROM

The form sends registration to Supabase Auth when a user submits it. The guide captures only the form switch, without submission.

HOW THE RESULT IS MADE

The user enters credentials and follows any required email confirmation. Returning to Sign In allows access after account creation.

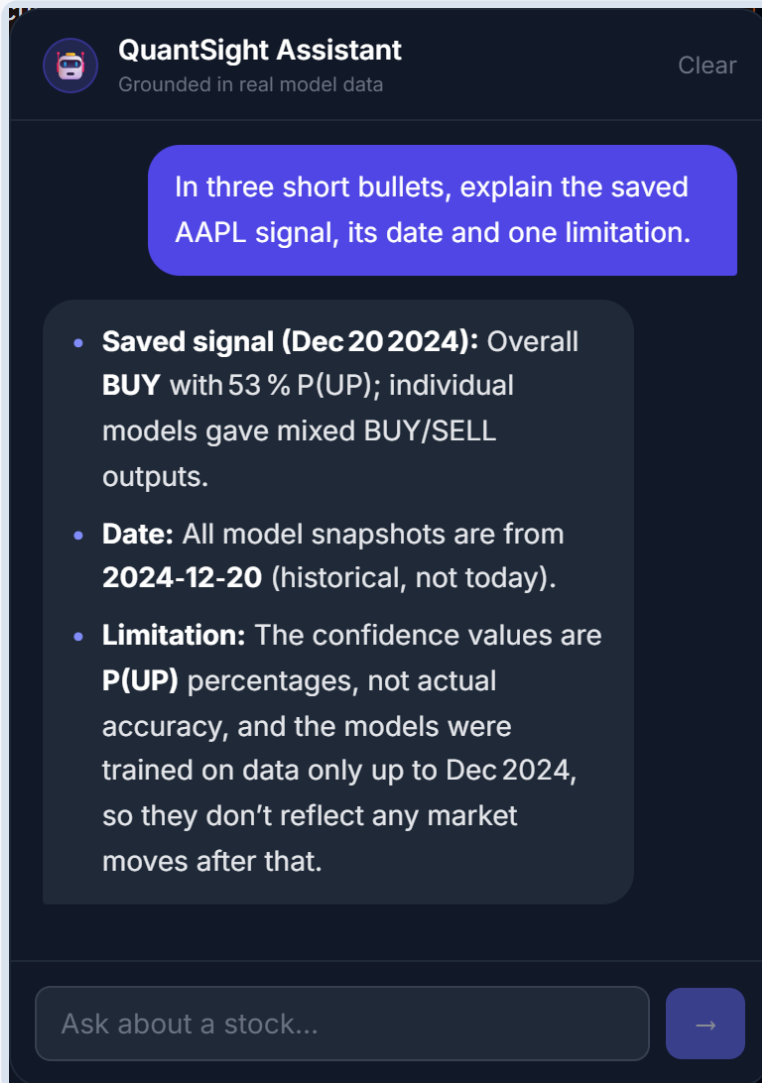
HOW TO READ IT

This guide does not verify a new account or its email delivery. Guest mode remains useful for a local demonstration without account changes.

REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: frontend/src/pages/Login.jsx](#)

AI Assistant: explain the project's evidence



REAL LOCAL SCREENSHOT / GUEST SESSION

[Code / evidence: backend/chatbot_engine.py](#)

WHAT IT IS

The floating assistant answers questions about indicators, saved stock signals, model results and project evidence.

WHERE DATA COMES FROM

The backend calls Groq, using its configured chat model, and exposes bounded read-only project tools. The current default is openai/gpt-oss-120b.

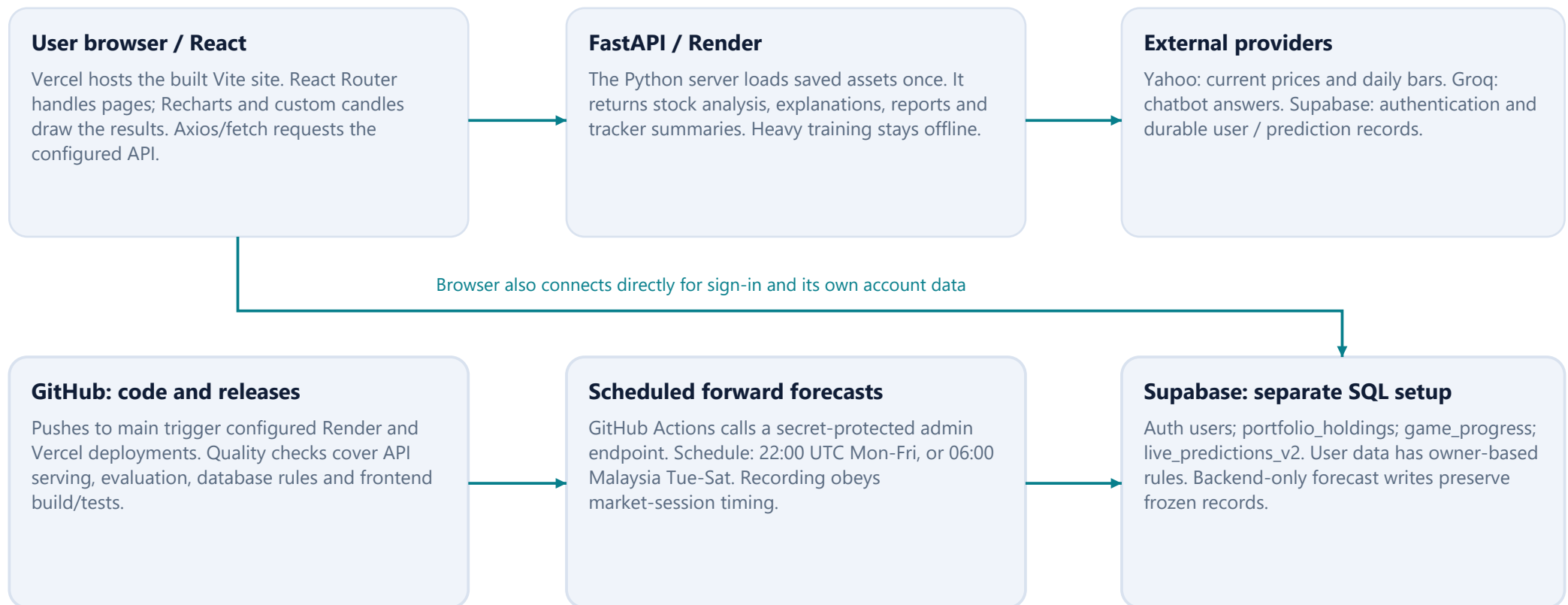
HOW THE RESULT IS MADE

The assistant can retrieve dated stock data before writing an answer. This captured real response describes AAPL's December 2024 signal and explains that confidence is not accuracy.

HOW TO READ IT

Generated text can still be wrong. Check dates and evidence. It needs internet, a valid backend Groq key and provider availability. The widget supports Stop, retry and Clear.

How the running services connect



Public addresses

quantsight.vercel.app

quantsight-rikc.onrender.com

Keep secrets on the backend

The Groq key, Supabase service-role key and admin-job secret belong in backend / hosting settings. The frontend uses the public Supabase client key. A git push does not apply SQL migrations or prove scheduled jobs succeeded.

How the simulation avoids looking ahead

1. Decide after a close

Use only the immediately previous session's saved forecast. Never select a stock using the execution day's signal.

2. Trade at the next open

Rebalance at the first observed session of each week. Rank eligible BUY scores; exclude SPY from stock selection.

3. Limit position size

Select up to five instruments. Target 1/5 of post-cost equity per slot, capped at 25%. Missing slots stay cash.

4. Carry shares and cash

Hold fractional shares continuously until a recorded change. Include gaps and weekends. Mark every session at its close.

5. Charge changed notional

Deduct 10 bps commission and 5 bps slippage on each buy or sell. Solve post-cost targets so the policy does not borrow.

6. Finish and compare fairly

Sell remaining holdings at the last close with exit costs. SPY and equal-weight benchmarks use matching dates and cost rules.

Audit: 15 portfolios and 462 sessions were independently replayed from their execution exports; reconstructed values matched within USD 0.000001.

Limits: no leverage, shorting, taxes, market impact or explicit dividend cashflows. Original adjustment and point-in-time universe provenance still need auditing.

[Code / evidence: docs/BACKTESTING.md](#)

Portfolio success: returns, risk and costs

Net return / CAGR

Net return = final equity / starting equity - 1. CAGR annualises growth using 252 sessions per year.

Read net costs and the observed dates. CAGR uses a trading-session convention.

Sharpe ratio

Mean daily excess return / sample standard deviation of excess returns x square root of 252.

Higher means more return per unit of variation in this sample. Default risk-free rate is 0%.

Sortino / Calmar

Sortino uses downside variation. Calmar = CAGR / absolute maximum drawdown.

They focus on loss variation or peak-to-trough damage; undefined denominators remain missing.

Drawdown / Duration

Drawdown = equity / earlier running peak - 1. Duration counts consecutive underwater sessions.

More negative means a deeper loss. Long recovery periods matter even with a good final return.

VaR95 / CVaR95

VaR95 is the observed daily loss threshold at the 5th return percentile. CVaR averages that lower tail.

These describe sample tail losses. They do not guarantee a maximum future loss.

Exposure / Turnover / Costs

Exposure is invested value / equity. Turnover tracks traded notional; costs track commissions and slippage.

These explain how active and costly a strategy is. Positive-return days are not winning trades.

Also in the backtest report: beta, descriptive annual alpha and information ratio versus SPY. No significance claim is made from these regression quantities.

[Code / evidence: docs/BACKTESTING.md](#)

Prediction quality: more than accuracy

Accuracy

Correct direction labels / all evaluated labels.

Compare with always-UP on the same rows; class imbalance matters.

Balanced accuracy / MCC

Balanced accuracy averages each class's recall. MCC measures agreement from -1 to 1.

Useful when a model predicts mostly UP. MCC near zero indicates little directional agreement.

Precision / Recall / F1

Precision: correct BUY / predicted BUY. Recall: detected UP / actual UP. F1 balances both.

Positive-class F1 differs from macro F1, which averages the two classes.

ROC-AUC / Average precision

AUC measures ranking across decision thresholds. Average precision summarises UP precision-recall.

Neither is a realised return. AUC near 0.5 indicates weak directional ranking.

Brier / Log loss

Brier = $\text{mean}((P(\text{UP}) - \text{outcome})^2)$. Log loss penalises confidently wrong probabilities.

Lower is better. They measure overall probability quality, not calibration alone.

Calibration / Coverage

Calibration compares predicted probabilities with observed frequencies. Coverage is the share retained.

Reliability bins can be noisy. High confidence with very low coverage may give a small, unrepresentative sample.

Also in the backtest report: beta, descriptive annual alpha and information ratio versus SPY. No significance claim is made from these regression quantities.

[Code / evidence: docs/BACKTESTING.md](#)

What can be shown, and what still needs evidence

Working system	Local screenshots cover all main routes, stock tabs, TreeSHAP, research reports, backtesting, guest tools and a real chatbot reply. The connected architecture uses GitHub, Render, Vercel and Supabase.
Live evaluation is still pending	The captured tracker has 44 logged and zero resolved predictions. A blank live accuracy is the correct result while outcomes are pending. One temporary database read failure recovered on retry.
Historical results are exploratory	The 2023-2024 period has influenced development. Fixed surviving tickers and uncertain original adjustment metadata limit the backtest. A stronger-looking historical model is not automatically a deployable winner.
Sentiment benefit is not established	Old GDELT news is dated 2026 and WSB ends in 2021; saved 2023-2024 text inputs are zero. Controlled news research has not established a robust probability advantage over simple baselines.
A fresh final test remains separate	The candidate 2025 snapshot is not scored in this guide. Its eligibility as an untouched final test depends on confirming whether those results were used before. Do not tune using the final test.
Personal and AI tools have limits	Portfolio daily P&L is an approximation; saved model signals can be older than a quote. Account views were reviewed without changing accounts. Chat answers require factual checks and provider access.

Describe the FYP as an auditable decision-support and evaluation system. Better prediction performance remains a research question to verify, not a claim to assume.

[Code / evidence: docs/IMPROVEMENT_ROADMAP.md](#)

Run the same interface on your computer

1. Backend - PowerShell window A

```
cd C:/Users/syeda/OneDrive/Desktop/quantsightv2/backend  
.venv/Scripts/python.exe -m uvicorn main:app --host 127.0.0.1 --port 8001
```

2. Frontend - PowerShell window B

```
cd C:/Users/syeda/OneDrive/Desktop/quantsightv2/frontend  
npm run dev -- --host 127.0.0.1 --port 3000
```

3. Open the dashboard

<http://127.0.0.1:3000> / Frontend .env: VITE_API_URL=http://127.0.0.1:8001

The local services were already running during capture. Use these commands after stopping them or restarting the computer; avoid launching duplicate servers on the same ports.

A useful demo order

Dashboard > Market > AAPL / TreeSHAP > AI Compare > Backtesting > Live Track Record > Portfolio / Game > Assistant.

Local mode avoids Render startup delay. Yahoo, Groq, sign-in and Supabase records still need internet. Saved charts and backtests read local assets.

[Code / evidence: HOW_TO_RUN.md](#)

How this guide was checked

Project baseline

- Documentation date: 5 October 2026, Malaysia. Code baseline: 8bd515a. Later releases may change screenshots, quotes and pending counts.
- Screenshots come from <http://127.0.0.1:3000> with the local API on port 8001. A new guest browser was used; no private account session was captured.
- Screenshots are cropped for readability. The narrow model-summary column is split into two side-by-side crops. The floating launcher is hidden during section captures; the actual chat panel is shown separately.
- A demonstration holding and one game answer were entered only in isolated guest storage. No broker orders, account changes, new live forecasts or final-test scoring were performed.
- The PDF was rendered page by page for visual review. The complete feature content is also saved in a readable Markdown companion.

Main implementation and evidence

[App routes and feature pages](#)

[API and serving paths](#)

[Feature construction and label](#)

[Data audit and coverage](#)

[Backtest method and formulas](#)

[Explainable AI and chatbot scope](#)

[Controlled finance research](#)

[Confidence versus coverage](#)

[Forward forecast protocol](#)

[Deployment schedule](#)

Interpretation rule

Keep four ideas separate: a language score, a model probability, observed direction accuracy and net portfolio performance. Each has its own data, date and measurement.